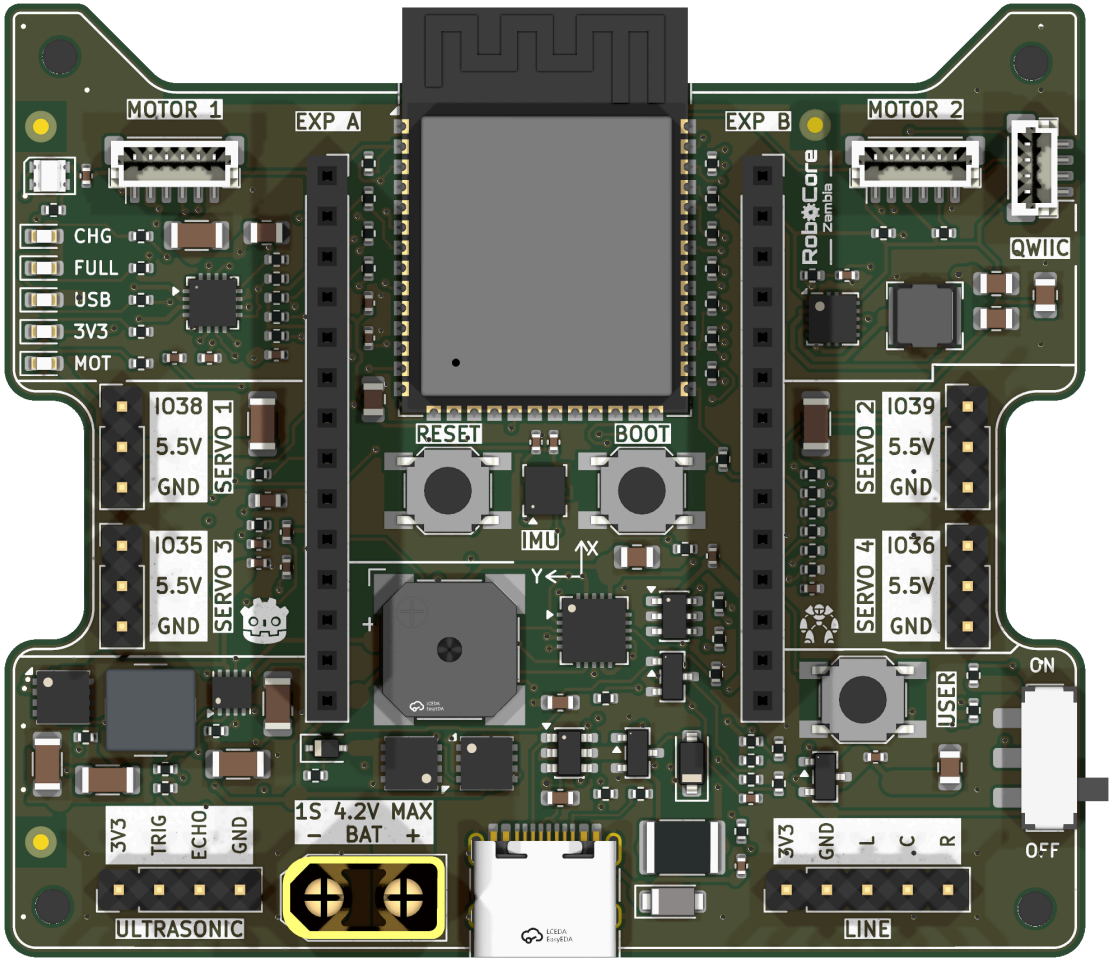
RoboCore V2 · robot controller, rev 2.0



AT A GLANCE

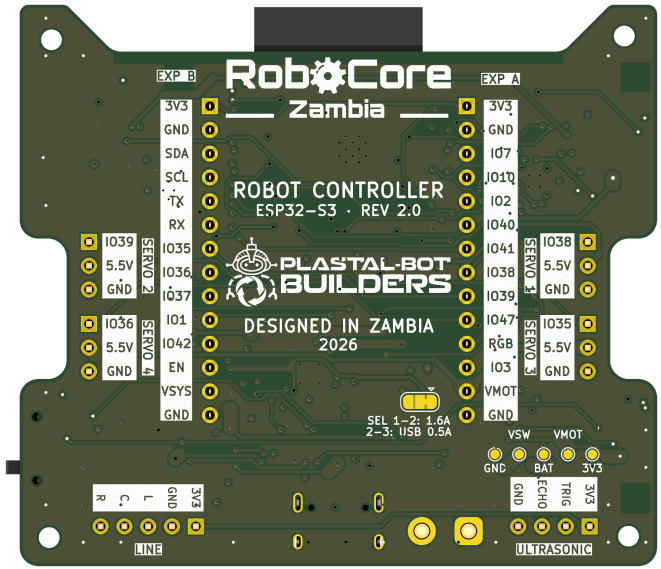
Board	68 × 60 × 1.6 mm, 4 layers, ENIG, green mask / white silk
Brain	ESP32-S3 (dual-core 240 MHz, 8 MB flash, 2 MB PSRAM)
Drives	2 DC motors with encoders, 4 servos, buzzer, RGB LED
Senses	IMU, 3-channel line follower, ultrasonic, battery voltage, motor current
Power	1S Li-ion pack (XT30), USB-C charging, 3.3 V + 5.5 V rails
Mounting	4 × M2 holes (2.2 mm), side notches for the servo cables
Design	Open-STEM XRP-compatible pinout, drawn from scratch in KiCad 10

MAIN PARTS

Function	Part	Note
MCU	ESP32-S3-WROOM-1-N8R2	Wi-Fi + BLE, native USB (one-click upload). Alternates N16R2, N8; never R8/R16V (octal PSRAM takes IO35–37)
Motor driver	TI DRV8411A (WQFN-16)	Both motors, 4 A peak; current limit 1.0 A (JST-SH contact rating)
Motor connectors	JST SH 6-pin ×2	XRP motor plug: M+, M–, 3.3 V, Enc A, Enc B, GND
IMU	ST LSM6DSOX (LGA-14)	6-axis; mounted at the board centre, over the wheel axle. Alternate LSM6DSO32
3.3 V rail	TI TPS63001 buck-boost	Up to ~1.2 A from a 1S cell, holds 3.3 V down to a flat cell
Motor / servo rail	TI TPS61089 boost → 5.5 V	SG90-type servos stay inside 4.8–6 V; Q102 P-FET cuts the boost when OFF
Charger	MCP73871 (USB-C)	1 A charge, power-path; charging works with the switch off. JP101 selects 1.65 A / 500 mA
Battery input	XT30 + reverse/over-voltage cut-off	1S Li-ion only (3.0–4.2 V). Q103/Q104 + TPS3700 survive a reversed or 2S/3S pack
Power switch	C&K JS102011 slide switch	Right edge; grounds the P-FET gates. OFF draws ~0.03 mA
USB-C	HRO TYPE-C-31-M-12 + ESD	Charging and programming
Lights	WS2812B-2020-V6 RGB + 5 status LEDs	CHG, FULL, USB, 3V3, MOT
Sound / buttons	CMT-8504 buzzer; RESET, BOOT, USER	PWM tones; USER on IO42
Ports	Qwiic (I2C), 2 × 14-pin expansion	3.3 V, VMOT, switched battery, spare GPIOs, UART, I2C

Top side — all SMD parts are on top; white knockout boxes carry the pin names of every header (servos, ultrasonic, line follower, battery)

Connectors and pinout



Bottom side — artwork, test pads, charge-current jumper JP101 and the pin names of both expansion headers

IR inputs use ADC1, which keeps working with Wi-Fi on. IMU axes: +X towards the top edge, +Y towards the left edge (firmware remaps).

CONNECTORS

Ref	Name	Type	Pins / notes
J301 / J302	Motor 1 / 2	JST SH 6	M+, M–, 3V3, ENC A, ENC B, GND (XRP order)
J401 J407 / J402 J408	Servo 1 3 / 2 4	2.54 mm 1×3 ×4	Signal, 5.5 V, GND — 1 + 3 on the left notch, 2 + 4 on the right
J403	Ultrasonic	2.54 mm 1×4	3V3, TRIG, ECHO, GND (HC-SR04 order: straight cable). 3.3 V module (HC-SR04+, RCWL-1601)
J404	Line follower	2.54 mm 1×5	3V3, GND, L, C, R — 3-channel TCRT5000-type module, analog outputs
J405 / J406	Expansion A / B	1×14 female ×2	See the pin order table
J201	Qwiic	JST SH 4	GND, 3V3, SDA, SCL
J102	Battery	XT30	1S Li-ion only, 4.2 V max; 1S2P/3P/4P packs need no change
J101	USB-C	16-pin	Charging + programming
JP101	Charge current	solder jumper (back)	1-2 adapter 1.65 A (default), 2-3 USB 500 mA
TP101–105	Test pads (back)		3V3, BAT, GND and rails

ESP32-S3 GPIO MAP

GPIO	Signal
IO0	BOOT button
IO1	IMU INT1
IO2	IR right (ADC)
IO3	Expansion (strapping pin)
IO4	Battery voltage (ADC, ÷2)
IO5 / IO6	Motor 1 / 2 current sense (ADC)
IO7	IR left (ADC)
IO8 / IO9	I2C SDA / SCL (IMU, Qwiic)
IO10	IR centre (ADC)
IO11 / IO12	Motor 1 IN1 / IN2
IO13 / IO14	Motor 2 IN1 / IN2
IO15 / IO16	Encoder 1 A / B

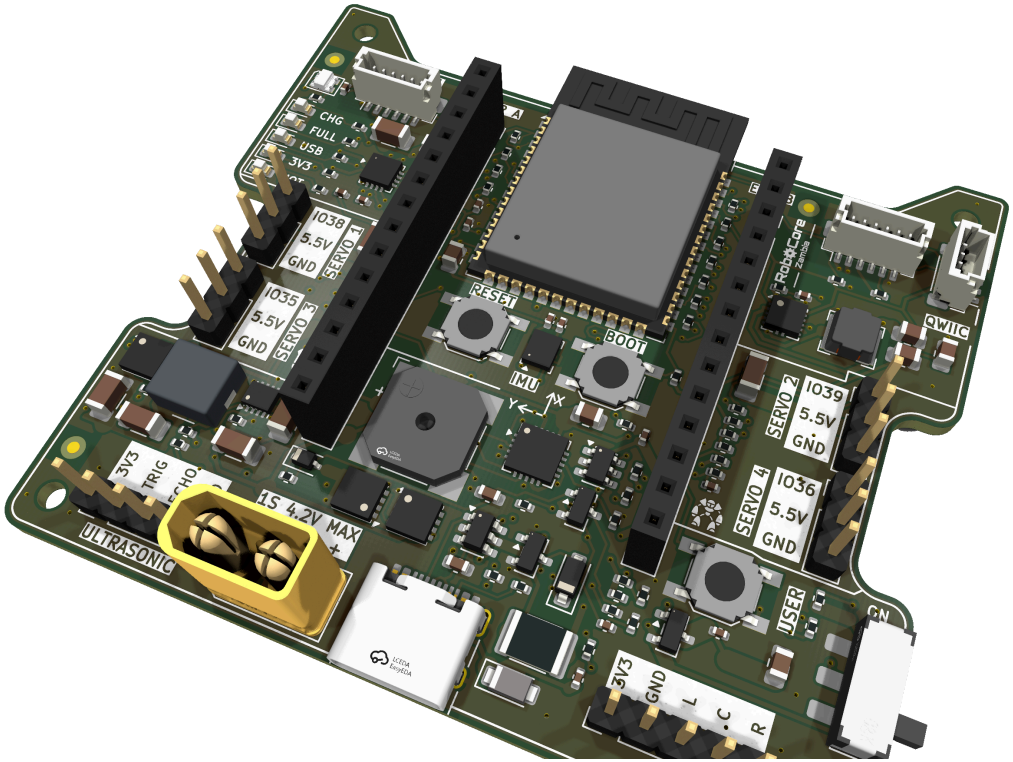
EXPANSION HEADERS (PIN 1 AT THE TOP)

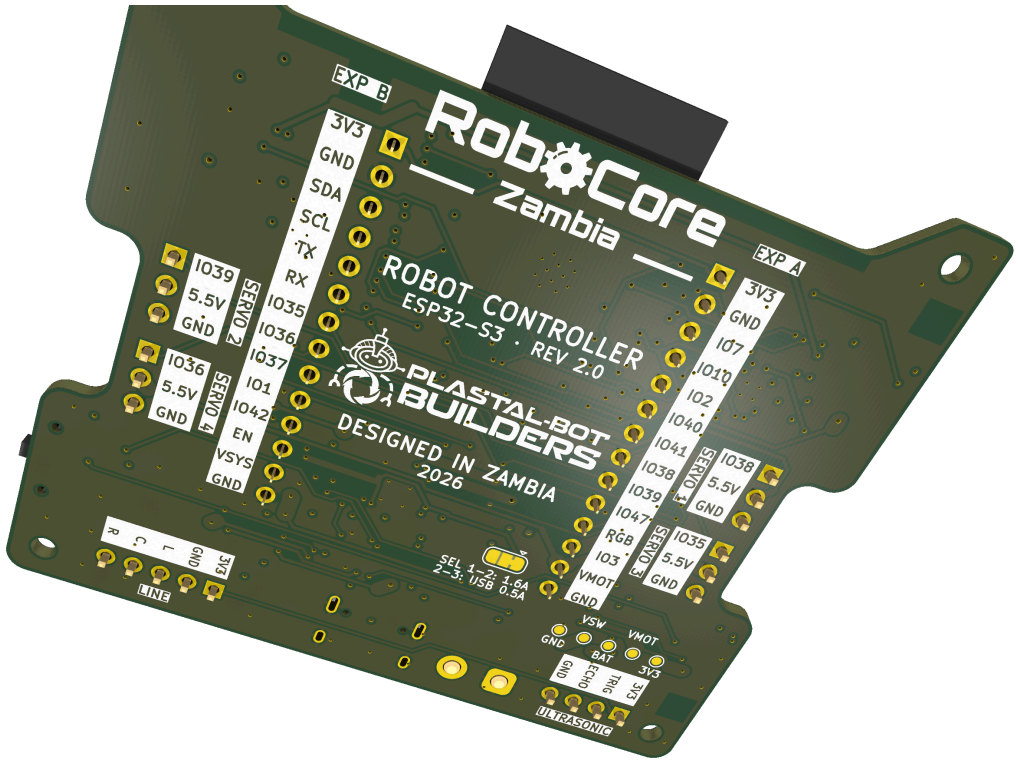
Pin	J405 EXPANSION A	J406 EXPANSION B
1	3.3 V	3.3 V
2	GND	GND
3	IR left (IO7)	I2C SDA (IO8)
4	IR centre (IO10)	I2C SCL (IO9)
5	IR right (IO2)	UART TX (IO43)
6	Ultrasonic TRIG (IO40)	UART RX (IO44)
7	Ultrasonic ECHO (IO41)	Servo 3 (IO35)
8	Servo 1 (IO38)	Servo 4 (IO36)
9	Servo 2 (IO39)	IO37
10	Buzzer (IO47)	IMU INT1 (IO1)
11	RGB LED data out	USER button (IO42)
12	IO3	EN (reset)
13	VMOT 5.5 V	VSYS_SW (switched battery)
14	GND	GND

GPIO MAP (CONTINUED)

GPIO	Signal
IO17 / IO18	Encoder 2 A / B
IO19 / IO20	USB D– / D+
IO21	Motor driver nFAULT
IO35 / IO36	Servo 3 / 4
IO37	Expansion B
IO38 / IO39	Servo 1 / 2
IO40 / IO41	Ultrasonic TRIG / ECHO
IO42	USER button
IO43 / IO44	UART TX / RX → expansion
IO47	Buzzer (PWM)
IO48	WS2812 RGB LED
IO45 / IO46	strapping pins, unused

3D views

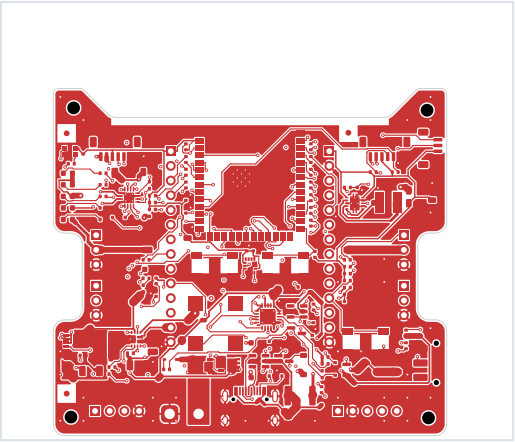




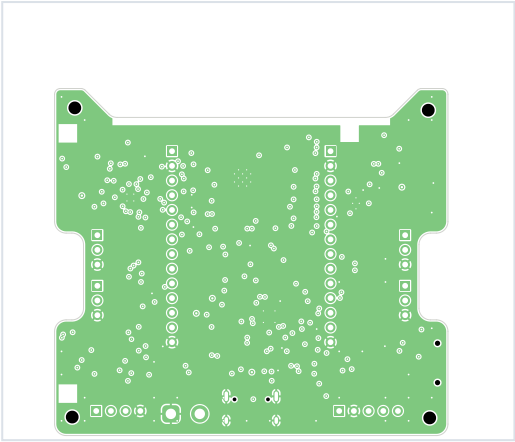
Top — ESP32-S3 at the top with the antenna over the edge, motor sockets upper left/right, servo headers in the side notches, expansion headers in the centre, sensors / battery / USB-C along the bottom edge, power switch on the right edge

Bottom — flat back (through-hole pin tails only, 1.5 mm): RoboCore Zambia and Plastal-Bot Builders logos, test pads TP101–105, JP101, expansion pin names

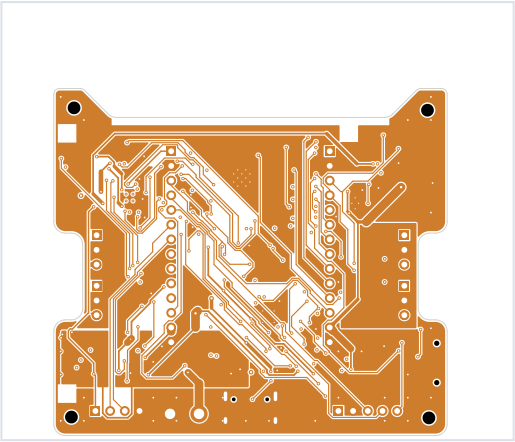
Copper layers · 1 : 1 plots



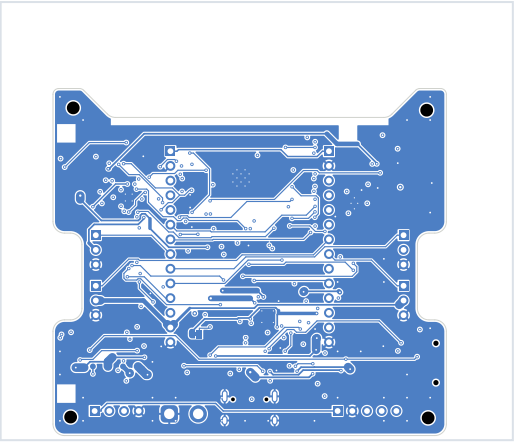
F.Cu — top signal layer



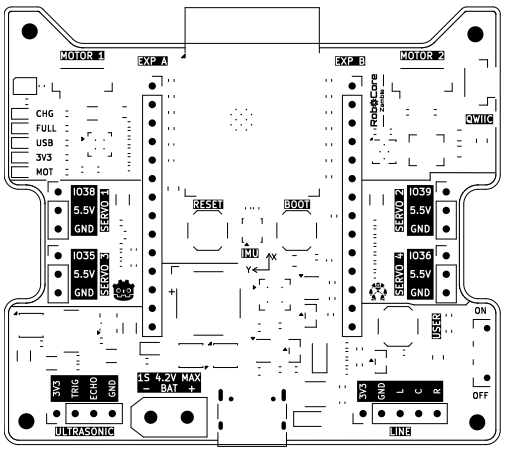
In1.Cu — inner 1, power / ground pours



In2.Cu — inner 2, signal and battery tracks



B.Cu — bottom signal layer



F.SilkS — front silkscreen

STACK-UP AND RULES

Item	Value
Layers	4: F.Cu · In1.Cu · In2.Cu · B.Cu, 1 oz copper outer and inner
Material	FR-4 S1000H TG150, 1.6 mm, green mask, white silkscreen, ENIG
Track / space	0.2 mm signals (6 / 6 mil rules); fat tracks for VMOT and the battery
Vias	0.3 mm hole / 0.6 mm ring, tented; 0.2 mm thermal vias in the IC pads (no resin fill)
Mask	no solder-mask web on the 0.15 mm gaps of U202 (LGA-14, 0.5 mm pitch)
Holes	4 × 2.2 mm non-plated (M2); side notches 5 mm deep for the servo cables
Artwork	front: knockout pin boxes beside every header, icons, dividers; back: logos and project text, 20 tented vias under it
Checks	DRC 0 errors, 0 unconnected; schematic parity clean (2026-10-08)

> Plots are at 1 : 1 as seen from the top (the bottom layer is not mirrored). Made with kicad-cli pcb export svg.

RoboCore V2 schematic · Root sheet

scale 1.04

RoboCore V2 — ESP32-S3 robot controller (68 x 60 mm, same outline as V1)

Parts follow the SparkFun XRP Controller where possible; circuit drawn independently.

Power

USB-C in, 1 A Li-ion charger with power path,
ON/OFF switch, 3.3 V buck-boost, 5.5 V boost (VMOT).

File: power.kicad_sch

MCU

ESP32-S3 module, buttons, RGB LED, buzzer,
LSM6DSOX IMU, I2C pull-ups + Qwiic.

File: mcu.kicad_sch

Motors

DRV8411A dual H-bridge with current sense,
two XRP motor connectors with encoders.

File: motors.kicad_sch

IO connectors

Servos, ultrasonic, line follower,
two 14-pin expansion headers.

File: io.kicad_sch

Power flow: USB-C 5 V -> charger -> VSYS -> ON/OFF -> VSYS_SW -> 3.3 V buck-boost and 5.5 V boost (VMOT).

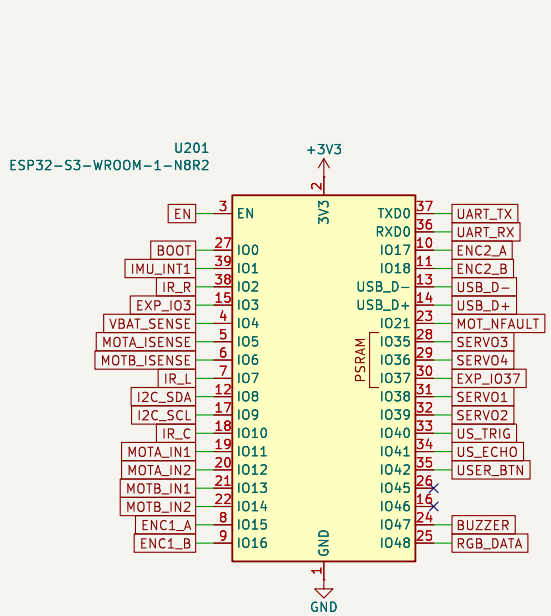
Battery: 1S Li-ion (cells in parallel) on XT30. Programming: USB-C straight to the ESP32-S3.

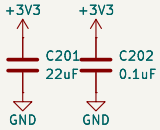
RoboCore V2 schematic · MCU · part 1 of 2 (top)

scale 0.95

ESP32-S3 MODULE

ESP32-S3-WROOM-1-N8R2: 8 MB flash, 2 MB quad PSRAM. USB D+/D- go straight to the module (upload + debug, no USB-UART chip). IO35-37 are free on N8R2 (quad PSRAM; IO35/36 drive servo 3/4); an R8 (octal PSRAM) module would use them. IO45/IO46 are strapping pins: left open.



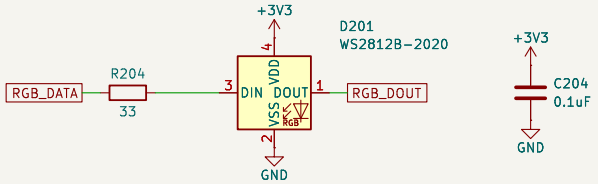


GPIO MAP

IO0 BOOT button
IO1 IMU INT1
IO2 IR right (ADC)
IO3 expansion
IO4 battery / 2 (ADC)
IO5, IO6 motor 1/2 current
IO7 IR left (ADC)
IO8, IO9 I2C SDA, SCL
IO10 IR centre (ADC)
IO11, IO12 motor 1 IN1, IN2
IO13, IO14 motor 2 IN1, IN2
IO15, IO16 encoder 1 A, B
IO17, IO18 encoder 2 A, B
IO19, IO20 USB D-, D+
IO21 motor nFAULT
IO35-37 expansion
IO38, IO39 servo 1, 2
IO40, IO41 ultrasonic
IO42 USER button
IO43, IO44 UART TX, RX
IO47 buzzer
IO48 RGB LED

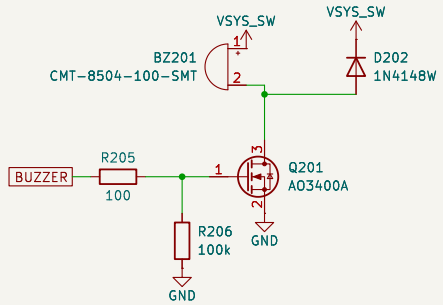
RGB STATUS LED

WS2812B-2020-V6 addressable RGB LED on IO48 (33R in series). V6 is rated 3.3-5.3 V (older 2020 versions need 3.7 V). DOUT goes to the expansion header to chain more LEDs.



BUZZER

Magnetic buzzer on VSYS_SW, switched by a low-side N-FET. A PWM tone on IO47 sets the pitch.
D202 absorbs the coil kick-back; R206 keeps the FET off while the ESP32 boots.



RESET / BOOT / USER BUTTONS

RESET (EN) restarts the ESP32. Hold BOOT (IO0) and tap RESET to force USB download mode.
USER (IO42) is free for your code. Each button pulls its line to GND; 10k pull-ups hold it high.

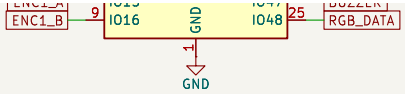
IMU (LSM6DSOX)

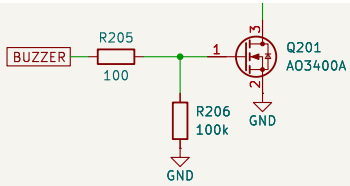
6-axis accelerometer + gyro on I2C, address 0x6A (SA0 = GND). CS high selects I2C. INT1 -> IO1.

+3V3 +3V3

RoboCore V2 schematic · MCU · part 2 of 2 (bottom)

scale 0.95





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RESET (EN) restarts the ESP32. Hold BOOT (IO0) and tap RESET to force USB download mode.
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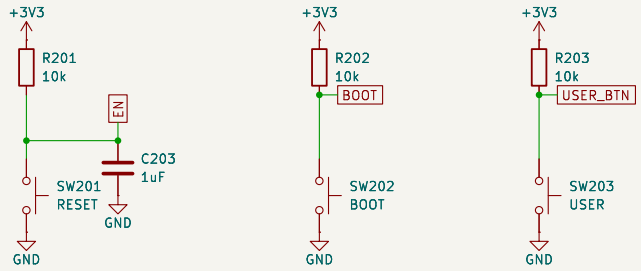


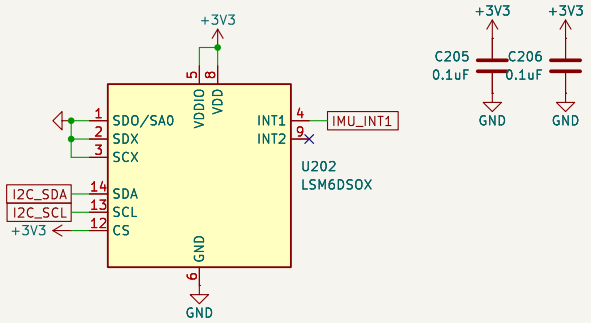
C203: reset delay

I2C BUS + QWIIC

4.7k pull-ups on SDA (IO8) / SCL (IO9). Qwiic socket (JST SH 4-pin, 3.3 V) for extra I2C sensors.

IMU (LSM6DSOX)

6-axis accelerometer + gyro on I2C, address 0x6A (SA0 = GND). CS high selects I2C. INT1 -> IO1.

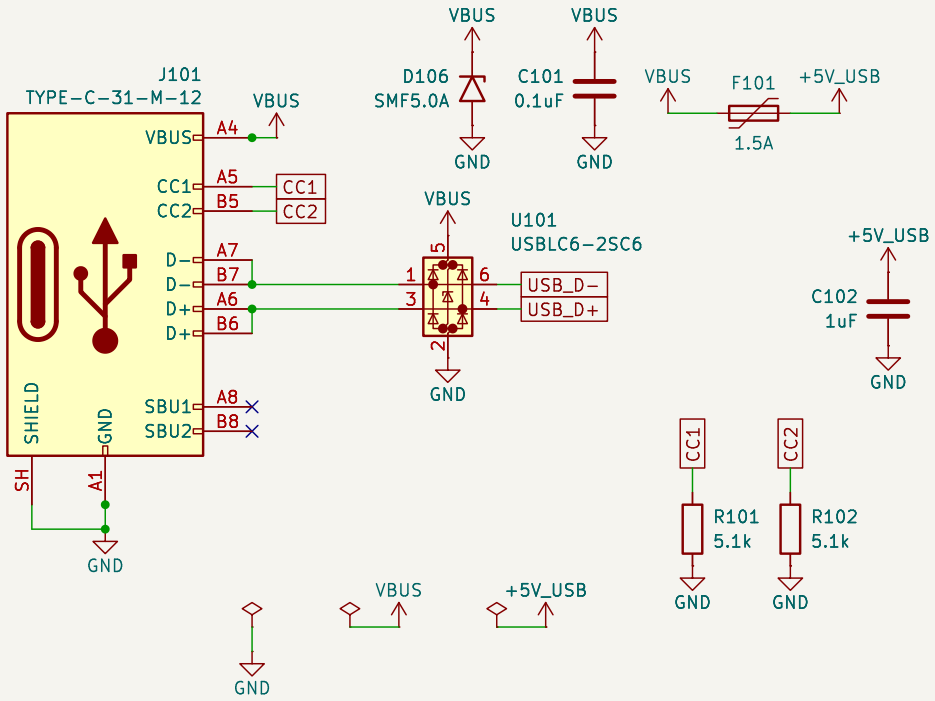


RoboCore V2 schematic · Power · part 1 of 4 (top left)

scale 1.28

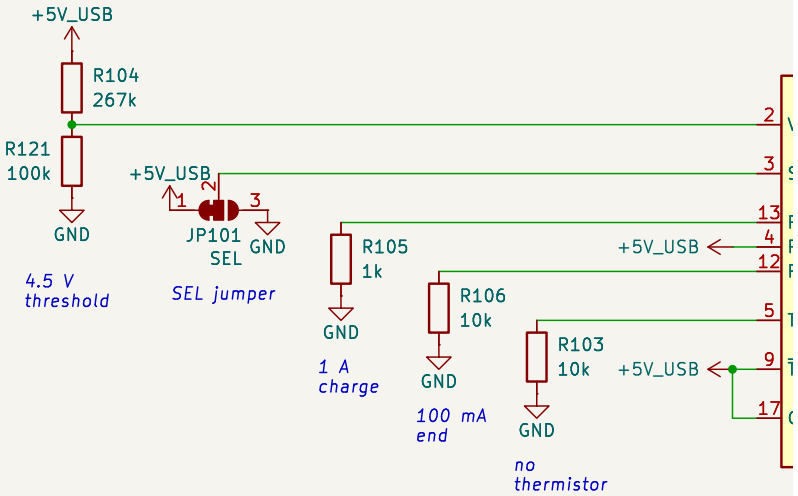
USB-C INPUT

USB-C socket for charging and programming. 5.1k on CC1/CC2 asks the source for plain 5 V.
USBLC6 clamps static on D+/D-, D106 clamps VBUS surges, and the 1.5 A resettable fuse protects the charger input.



BATTERY CHARGER + POWER PATH (MCP73871)

Charges the 1S Li-ion pack (4.20 V) at up to 1 A and powers the robot (VSY
VPCC divider: charging backs off if USB sags below 4.5 V. JP101 SEL: 1-2 =



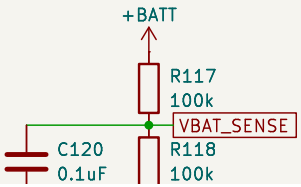
TE, CE, PROG2 tied high:

ON/OFF SWITCH

Switch ON grounds the P-FET gates: Q101 passes VSYS to VSYS_SW (3.3 V, buzzer, boost enable) and Q102 connects the pack to the boost.
Switch OFF: R110 holds both gates high, so everything including VMOT is off (charging still works).

BATTERY VOLTAGE SENSE

100k / 100k halves the pack voltage for the ESP32 ADC (IO4).



POWER LEDS

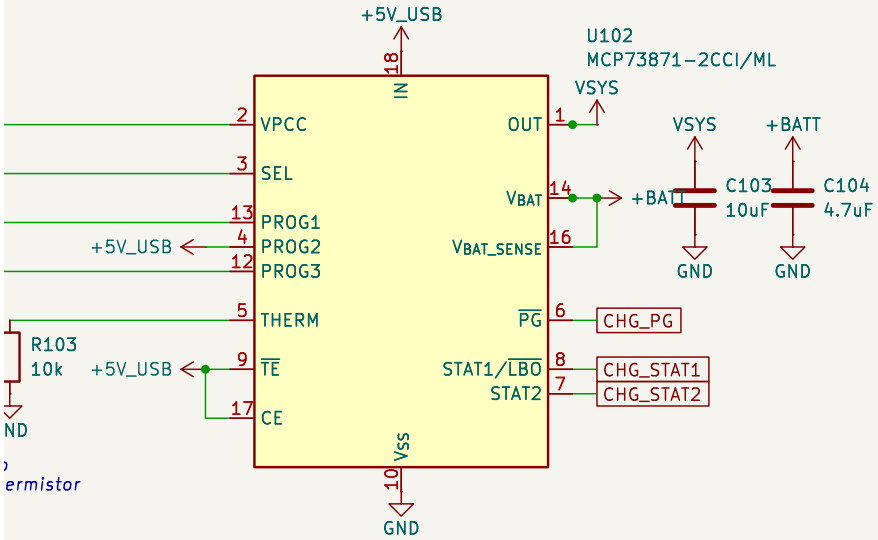
Green = 3.3 V rail on. R
rail (VMOT) on.

RoboCore V2 schematic · Power · part 2 of 4 (top right)

scale 1.28

'3871)

d powers the robot (VSYS) at the same time, from USB or the pack.
.5 V. JP101 SEL: 1–2 = adapter mode 1.65 A (default), 2–3 = USB 500 mA.

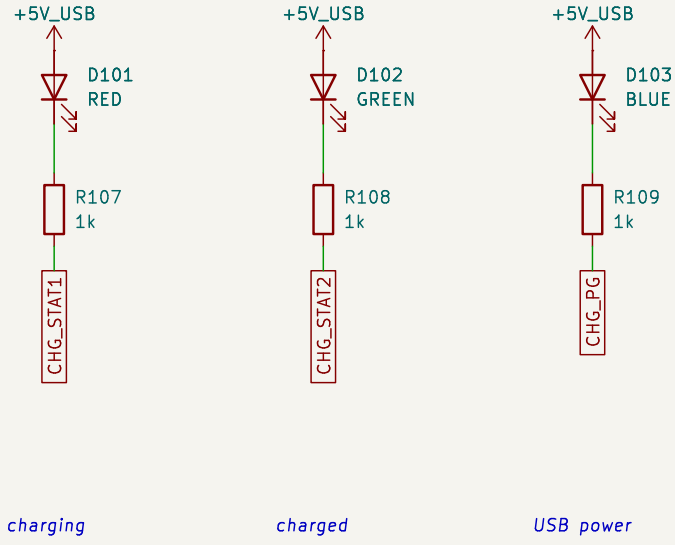


TE, CE, PROG2 tied high: safety timer off (big pack), charger on.

+BATT: battery after the input protection

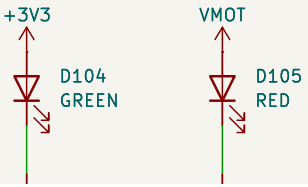
CHARGE STATUS LEDS

Driven by the charger: red = charging, green = charge complete, blue = USB power present.



'OWER LEDS

reen = 3.3 V rail on. Red = motor il (VMOT) on.



TEST POINTS

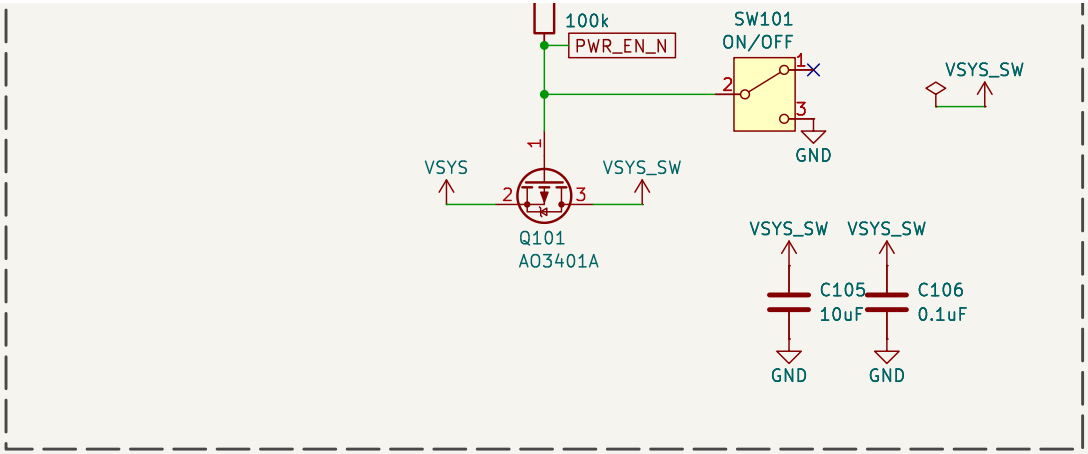
1 mm pads for bring-up measurements.

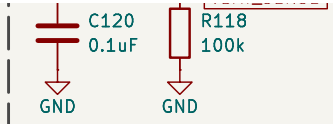
MECHANICAL

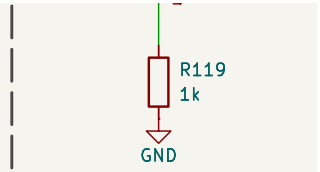
M2 mounting holes (same places as V1) and fiducials for assembly.

RoboCore V2 schematic · Power · part 3 of 4 (bottom left)

scale 1.28

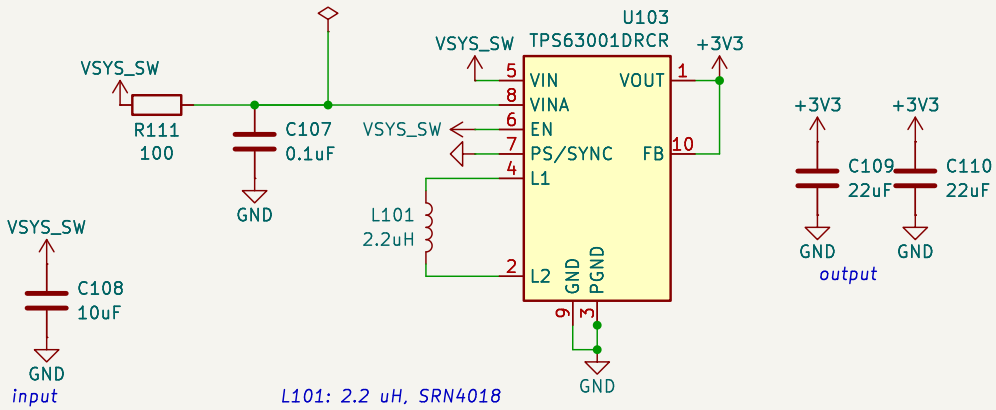






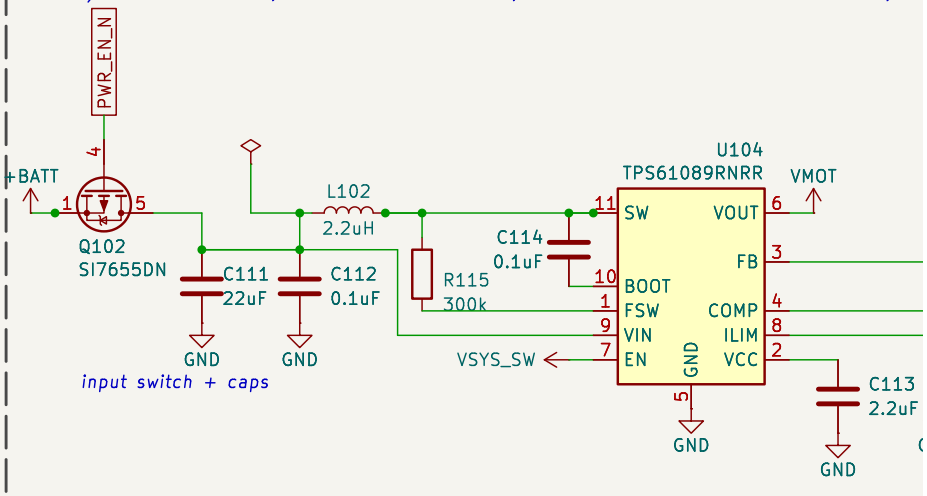
3.3 V REGULATOR (TPS63001 BUCK–BOOST)

Fixed 3.3 V from VSYS_SW (1.8–5.5 V in): holds 3.3 V from a full cell, a flat cell or USB.
R111/C107 filter the control supply (VINA). PS/SYNC low = power-save mode at light load.



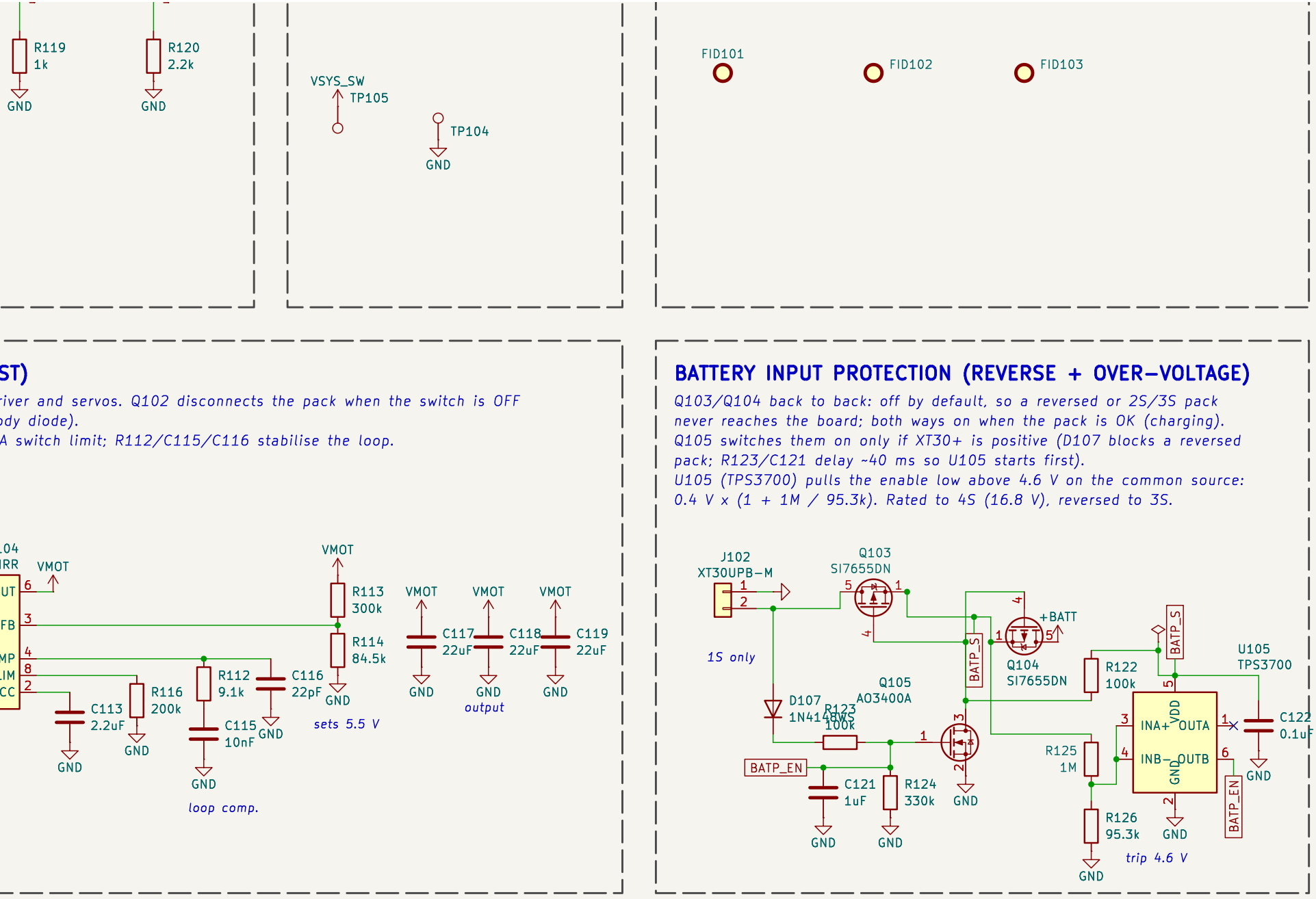
5.5 V MOTOR / SERVO SUPPLY (TPS61089 BOOST)

Boosts the pack (3.0–4.2 V) to VMOT = 5.5 V for the motor driver and servos. (a disabled TPS61089 still passes VBAT to VMOT through its body diode).
R113/R114 set 5.5 V; R115 sets ~500 kHz; R116 sets a 5.15 A switch limit; R1



RoboCore V2 schematic · Power · part 4 of 4 (bottom right)

scale 1.28

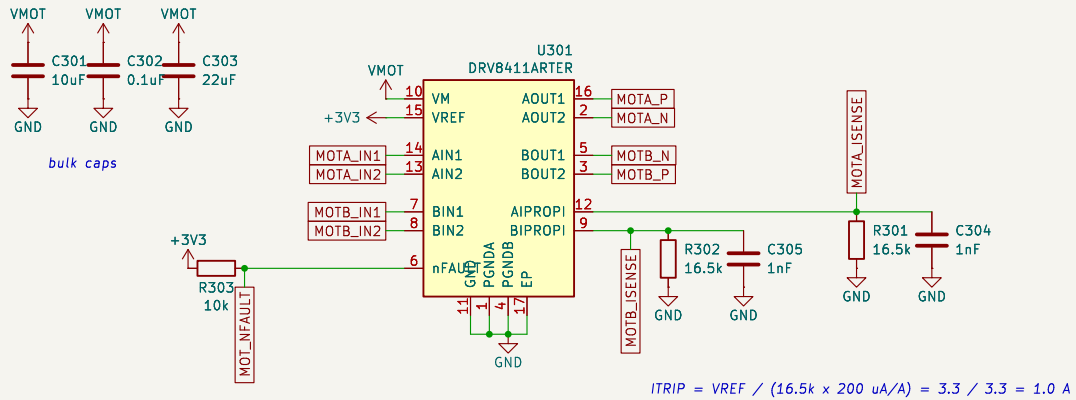


RoboCore V2 schematic · Motors

scale 0.97

DUAL MOTOR DRIVER (DRV8411A)

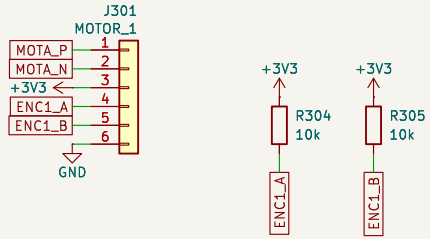
Two H-bridges powered from VMOT (5.5 V). Per motor, PWM on IN1/IN2 sets speed and the pair sets direction; all low = coast + sleep. IPROPI gives a copy of the motor current (200 uA per A) into 16.5k: the ESP32 reads it, and the driver limits each motor to 1.0 A (JST-SH contact rating).



ITRIP = VREF / (16.5k x 200 uA/A) = 3.3 / 3.3 = 1.0 A

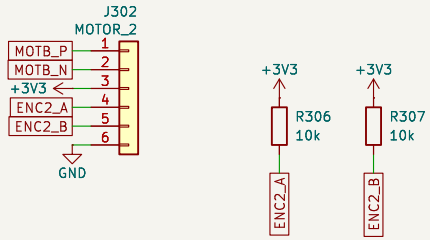
MOTOR 1 (XRP JST-SH 6-PIN)

XRP motor plug: 1 M+, 2 M-, 3 3.3 V, 4 encoder A, 5 encoder B, 6 GND.
10k pull-ups on the encoder outputs.



MOTOR 2 (XRP JST-SH 6-PIN)

Same plug as motor 1.
10k pull-ups on the encoder outputs.

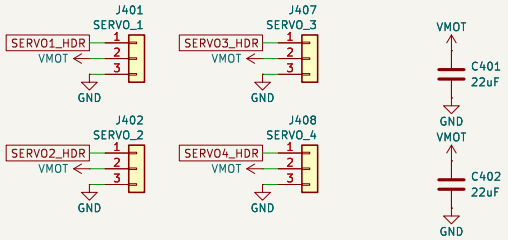


RoboCore V2 schematic · IO connectors

scale 0.82

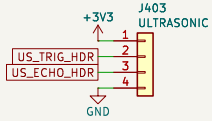
SERVOS

Four hobby-servo headers (signal, +, −) as on the XRP. Power is VMOT (5.5 V); C401/C402 are local bulk storage.
Servo 1/2 on IO38/IO39, servo 3/4 on IO35/IO36 (all four signals are also on the expansion headers).



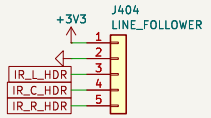
ULTRASONIC SENSOR

HC-SR04-type ultrasonic module on a 4-pin 2.54 mm header: 3V3, TRIG, ECHO, GND (the HC-SR04 pin order, so a straight cable works). Use a 3.3 V-capable module (HC-SR04+, RCWL-1601). Header again since 2026-10-08.



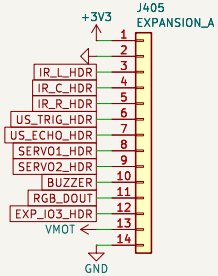
LINE FOLLOWER (3-CHANNEL IR)

3-channel IR line-follower module (TCRT5000 type) on a 5-pin 2.54 mm header: 3V3, GND, L, C, R.
Analog outputs to ADC1: IO7 left, IO10 centre, IO2 right. Header again since 2026-10-08.



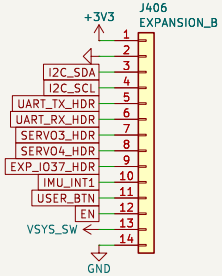
EXPANSION HEADER A

Female 2.54 mm header: the sensor, servo, buzzer and LED signals, for prototyping.



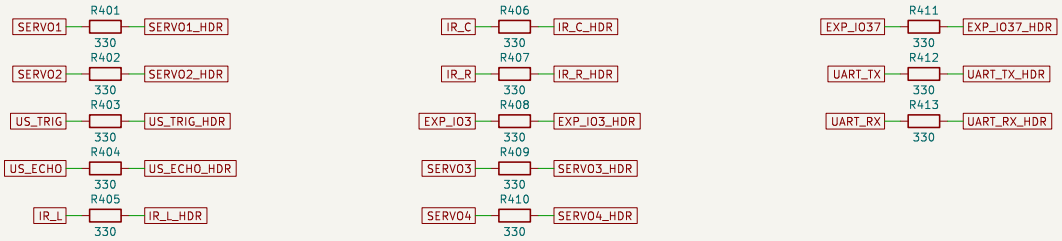
EXPANSION HEADER B

Female 2.54 mm header: I2C, UART0, servo 3/4 signals (IO35/IO36), spare IO37, IMU interrupt, USER button, EN and VSYS_SW.



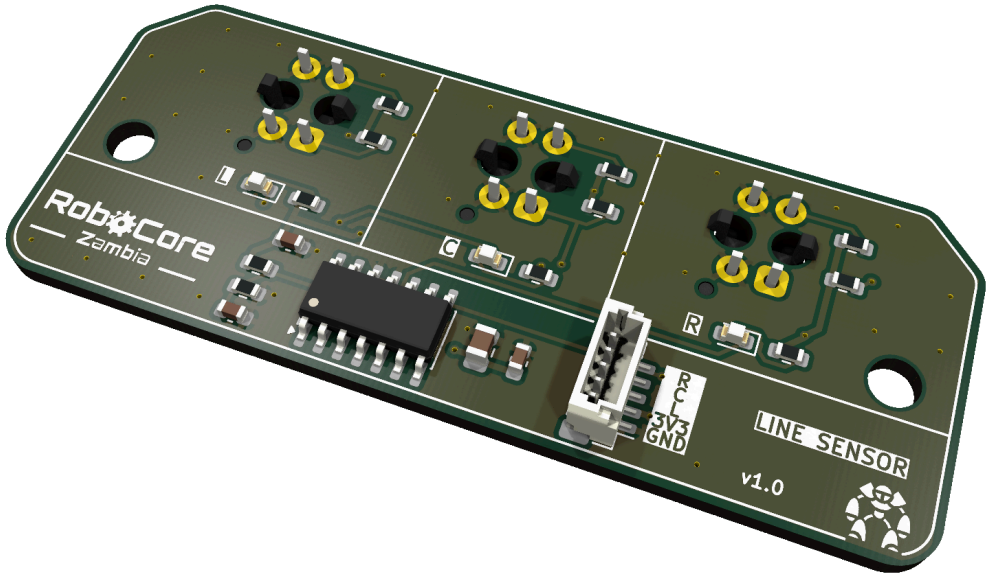
HEADER PROTECTION

330R in series with every ESP32 pin that leaves the board: limits the current if a pin is shorted to a rail or zapped.
Nets ending in _HDR are the connector side.

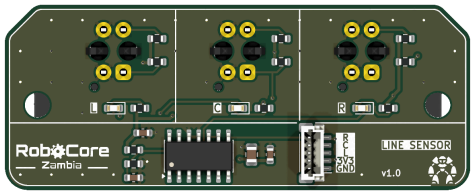


Line sensor and ultrasonic · our own versions of the two bought-in modules

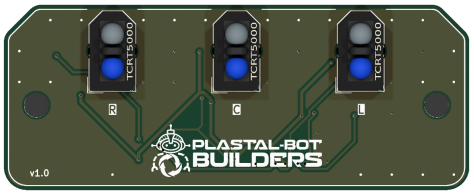
LINE SENSOR 60 × 24 MM, 2 LAYERS



3 × TCRT5000 on the underside, LM339 comparator, 3 green indicator LEDs



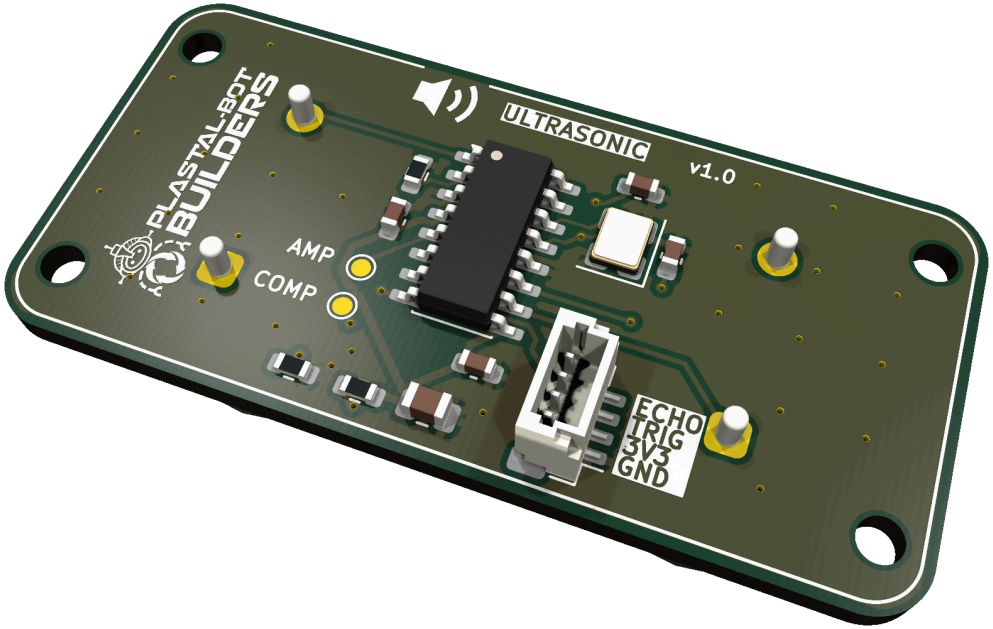
top



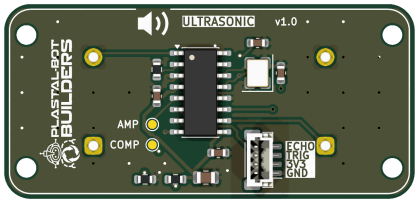
underside (mirrored silkscreen)

Item	Value
Connector	JST SH 5: GND, 3V3, L, C, R (analog, 0.3 V white … 2.8 V black)
Supply	3.3 V, about 45 mA
Mounting	2 × M3, 51.6 mm apart; underside 9–13 mm above the floor
Sensors	16 mm apart, for 19 mm tape; LEDs show which channel sees the line

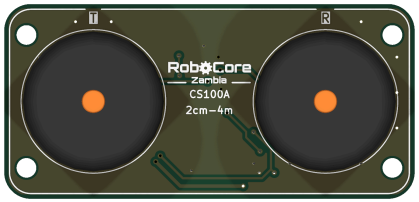
ULTRASONIC 46 × 22 MM, 2 LAYERS (HC-SR04 SIZE)



CS100A ranging chip, 8 MHz crystal, 40 kHz transmit + receive transducers



rear face: chip, crystal, connector



front face: transducers, T left / R right

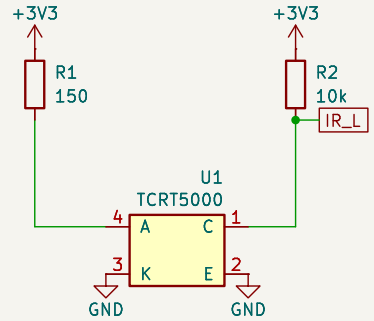
Item	Value
Connector	JST SH 4: GND, 3V3, TRIG, ECHO (HC-SR04 timing: 58 µs per cm)
Supply	3.3 V, 5.3 mA
Mounting	4 × M2 at (±20.5, ±7.5) mm; ≥ 4–5 cm above the floor (70° beam)
Range	2 cm to about 4 m; T left, R right seen from the front

Line sensor schematic · Line sensor

scale 1.24

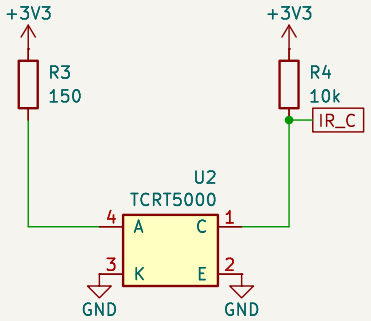
LEFT SENSOR

IR LED ~14 mA (150 from 3.3 V). Output IR_L: low over white, high over the line.



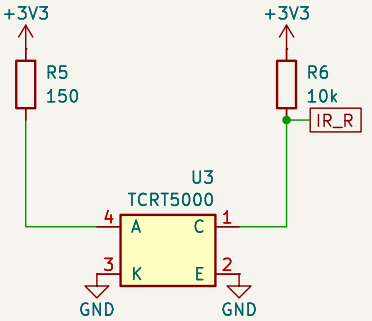
CENTRE SENSOR

IR LED ~14 mA (150 from 3.3 V). Output IR_C: low over white, high over the line.



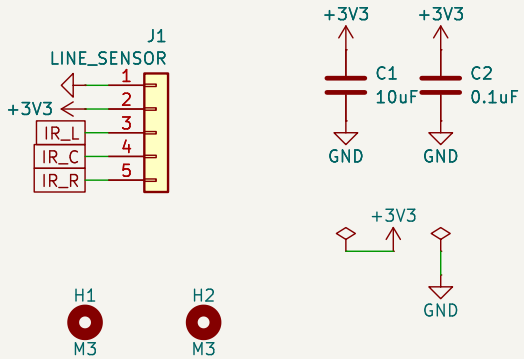
RIGHT SENSOR

IR LED ~14 mA (150 from 3.3 V). Output IR_R: low over white, high over the line.



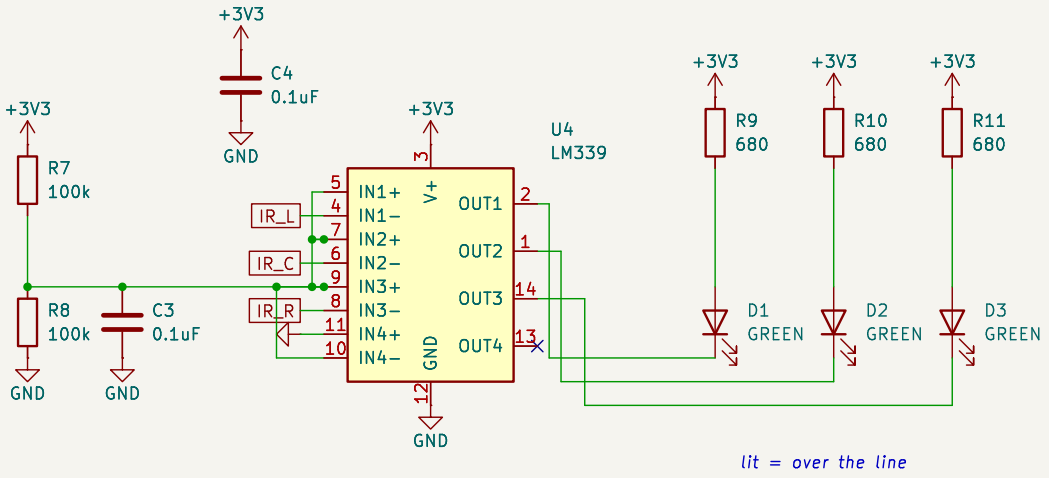
CONNECTOR

JST SH 5-pin to the RoboCore V2 (J404): GND, 3V3, L, C, R. Same pin order as the board's socket.



LINE INDICATORS (LM339)

Each comparator sinks its LED when the sensor voltage is above VREF (1.65 V), i.e. over the black line.
VREF = half of 3.3 V. The spare comparator has its inputs tied so its output stays low (pin 13 unused).

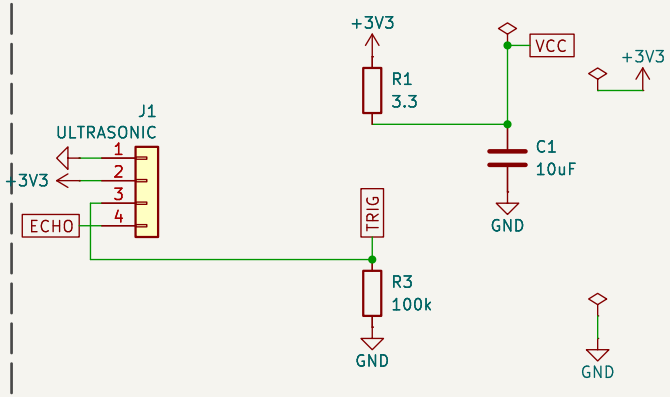


Ultrasonic schematic · Ultrasonic

scale 1.17

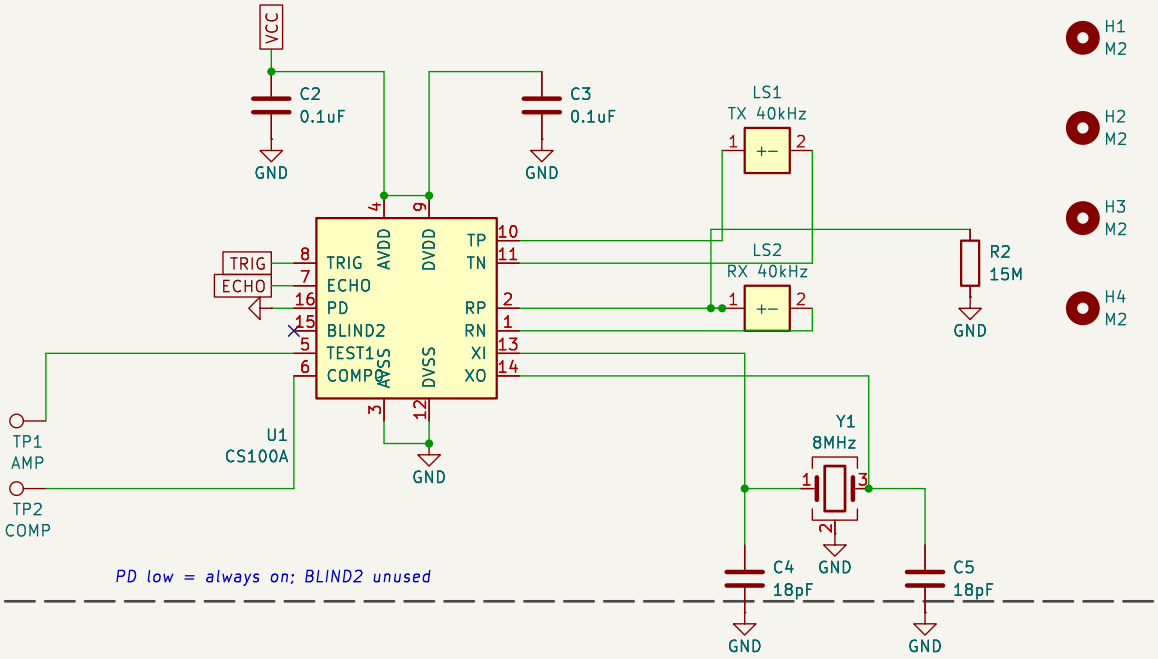
CONNECTOR + SUPPLY FILTER

JST SH 4-pin to the RoboCore V2 (J403): GND, 3V3, TRIG, ECHO (Qwiic power order).
R1/C1 keep the transmitter's current pulses off the robot's 3.3 V rail (CS100A reference circuit).

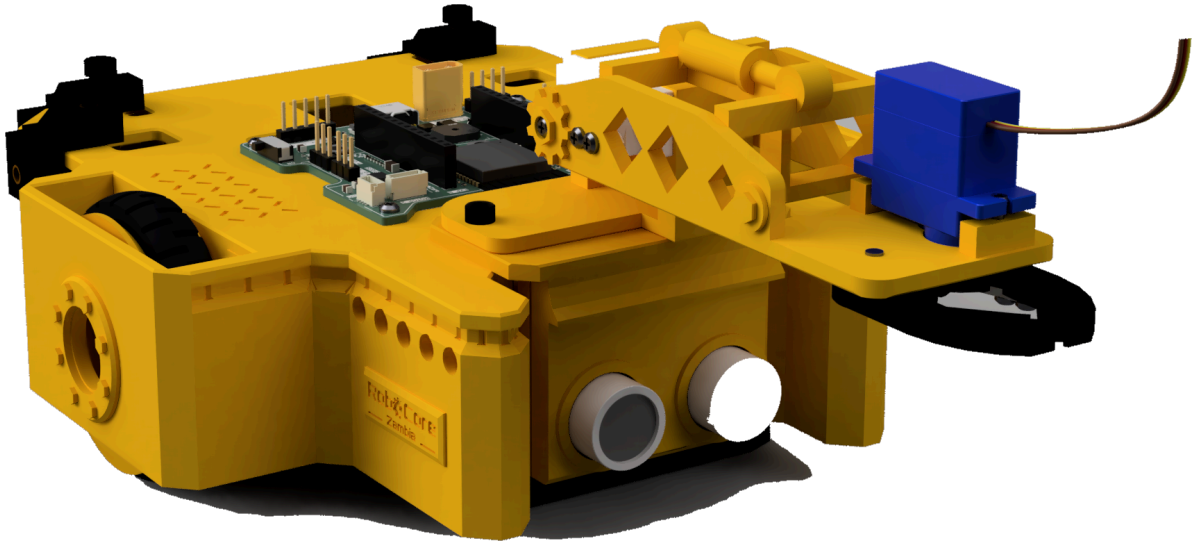


RANGING (CS100A)

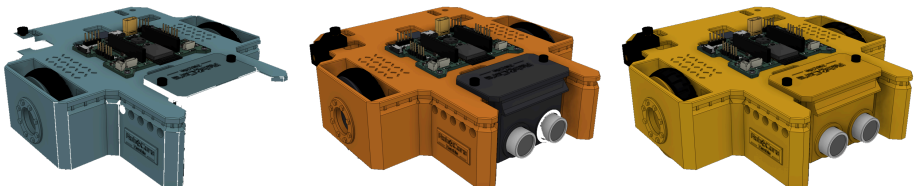
Y1 8 MHz crystal (12 pF load, 18 pF caps). R2 15M sets the receiver sensitivity: lower it (5.1M–22M) if far readings jump around; raise it for more range. TP1/TP2: amplifier and comparator outputs, for tuning.



RoboCore V2 robot · chassis rev 5 (Fusion 360)



Rev 5 with the arm module — construction-yellow scheme. Deck-down print, no supports; motors snap in from below; battery slides in from the back



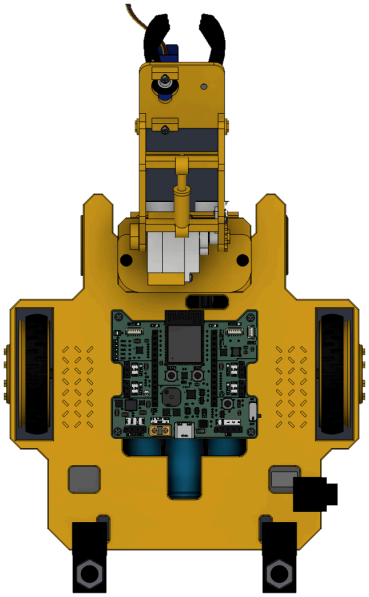
Colour schemes: teal, orange, construction yellow

KEY NUMBERS

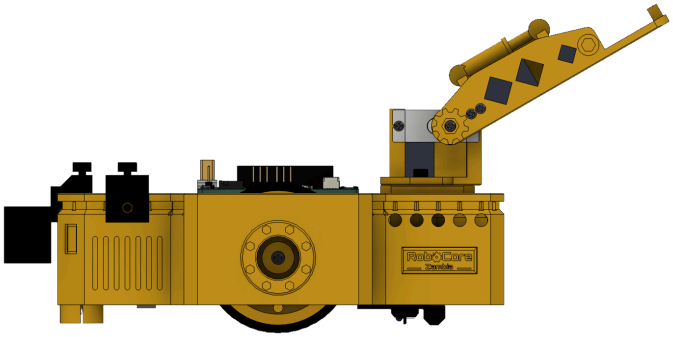
Item	Value
Body	162 × 166 mm over the walls, 52 mm to the deck top; cross-shaped plan, every corner chamfered 45°
Drive	2 × TT gear motors with encoders (6 V, 160 rpm), wheels clamp to the shaft; twin ½" ball casters at the rear
Board	RoboCore V2 on the deck, 4 × M2 heat-set inserts; IMU over the wheel axle
Battery	BatteryHolder3S, 3 × 18650 wired 1S3P, slides in from the back behind two snap flaps
Sensors	HC-SR04 in a headlight-style pod on the front dock; line-sensor sled under the nose (3 heights)
Arm	excavator-style boom, MG90S shoulder, SG90 gripper; bolts to the front dock with 2 × M3
Expansion	flush bus groove with 54 detents, 10 Technic-pitch holes, M3 dock inserts, 8 mm grid
Printing	chassis deck-down with no supports, ≥ 20 % infill; every file in print/ is already oriented

WHAT IT FIXES (VS. THE XRP)

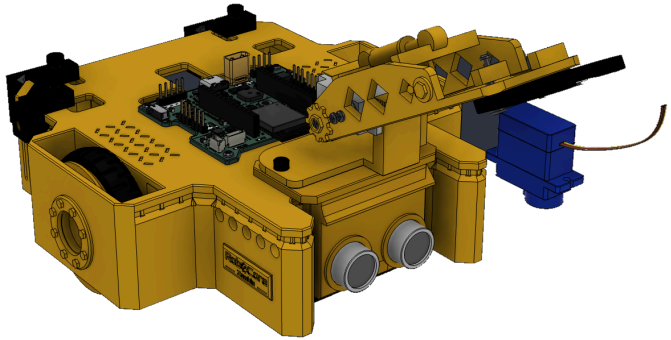
- Real tyres with 2 mm frame clearance, wheels screwed to the shaft, not press-fit
- IMU directly over the axle and a rigid 4-screw board mount: no drift from flex
- Li-ion pack with USB-C charging and reverse / over-voltage protection instead of AA cells
- Motors held by snap hooks and stops, swappable in minutes; screws + inserts where students take things apart
- Antenna left in the open at the top edge



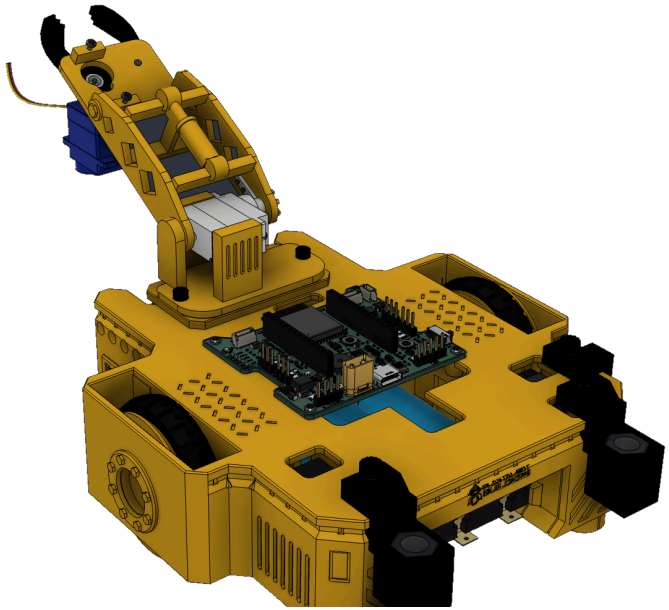
Top



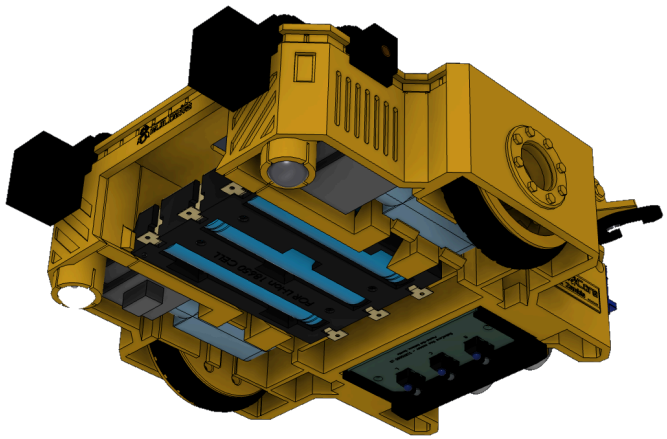
Side (arm raised)



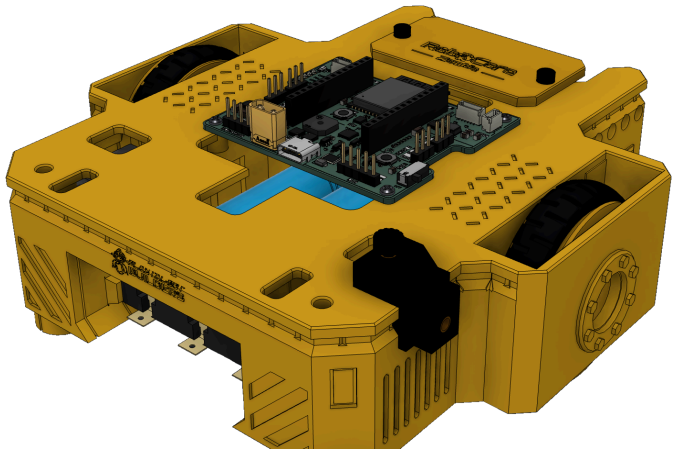
Front isometric



Rear isometric — battery door, cable ports, ballast clips

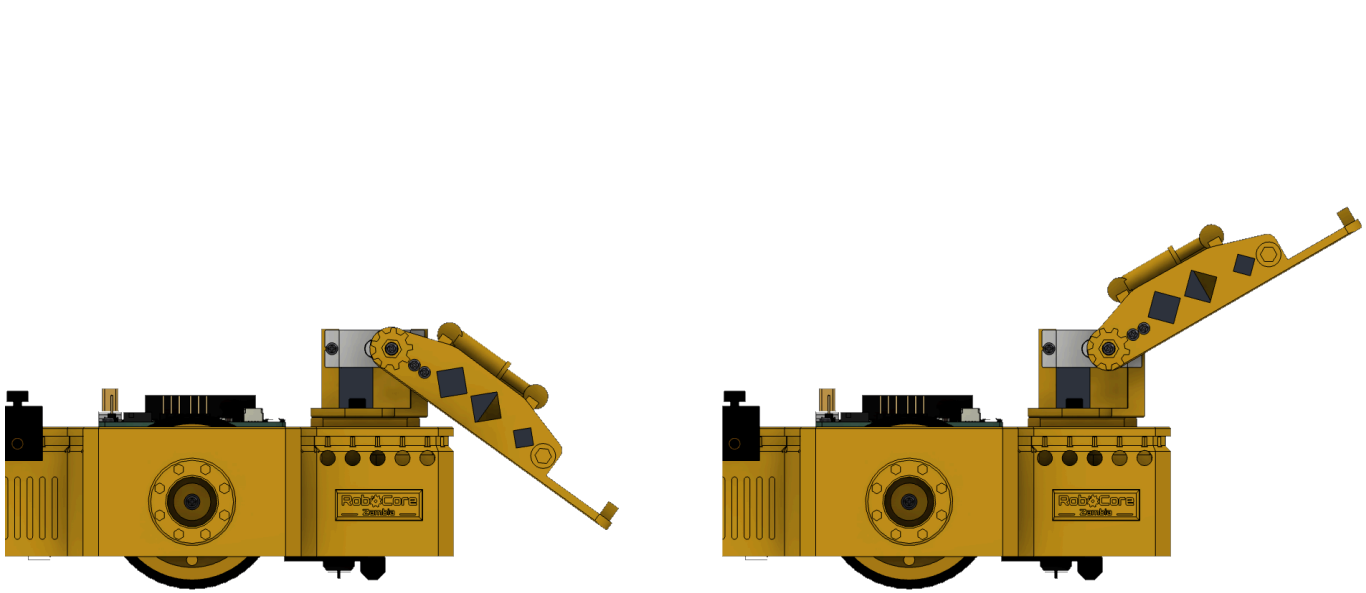


Underside — open body, motors, line-sensor sled, casters

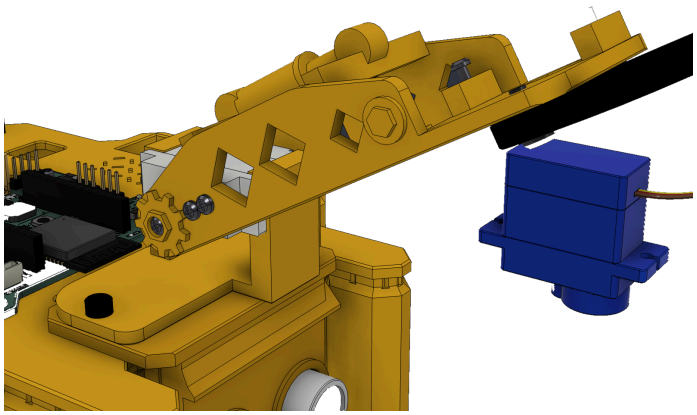


Without the arm — the base robot

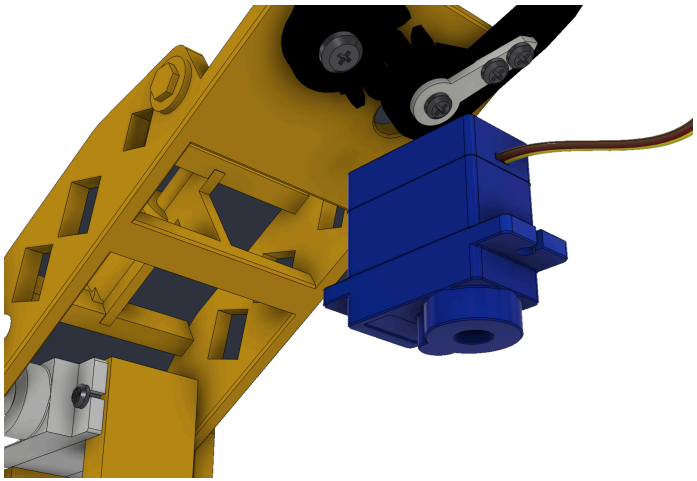
Arm module · excavator-style



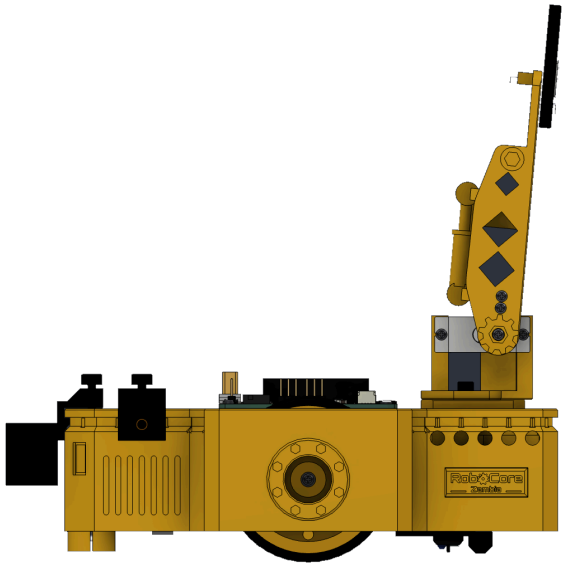
Arm poses — on the floor, carrying, raised



Shoulder — MG90S servo, RoboCore gear badge, hex bolt head, truss windows, hydraulic cylinder

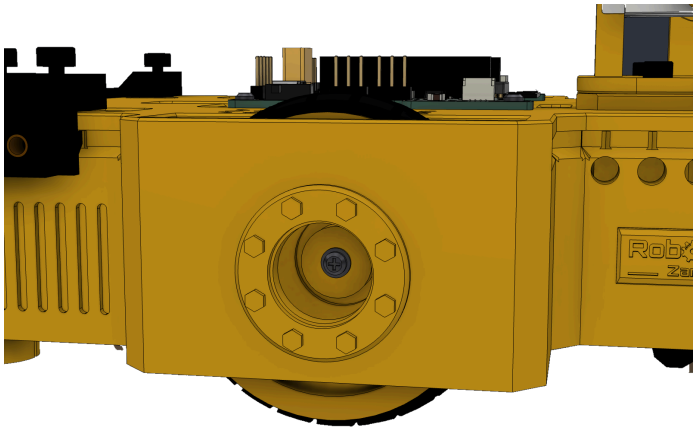


Gripper — SG90 servo, horn and screws under the wrist; gear-linked jaws with bucket teeth

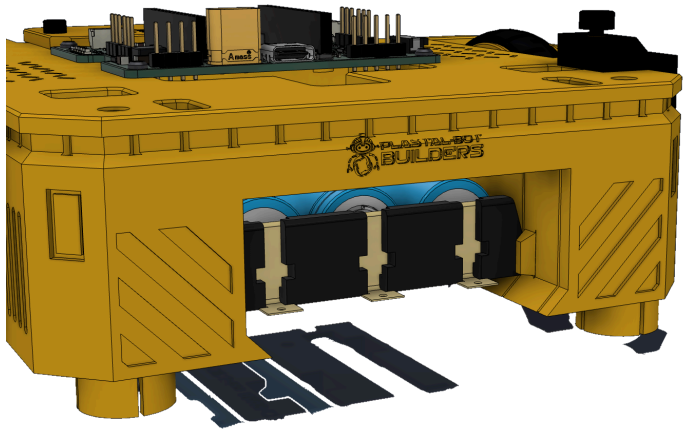


Reach — arm lowered to the floor

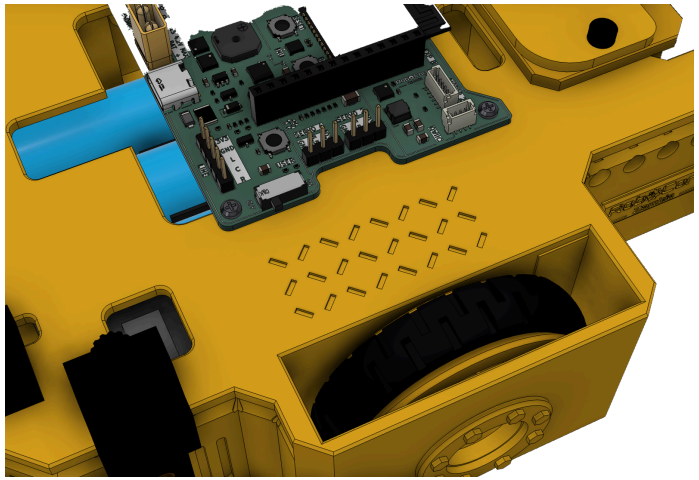
Chassis details · rev 5



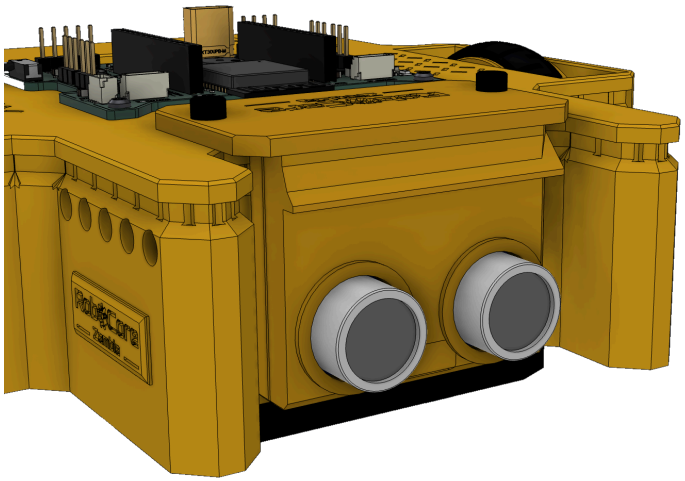
Wheel side — hub, fender arch, engine louvres, RoboCore badge, Ø16 access hole for the wheel screw



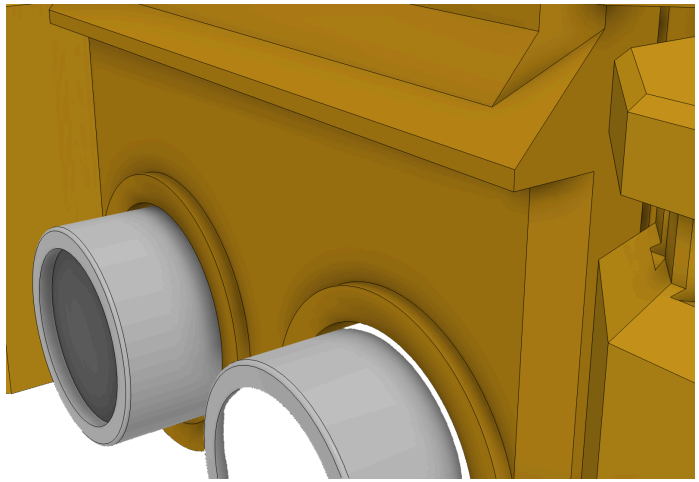
Rear — hazard chevrons, tail lights, battery opening with snap flaps, cable ports



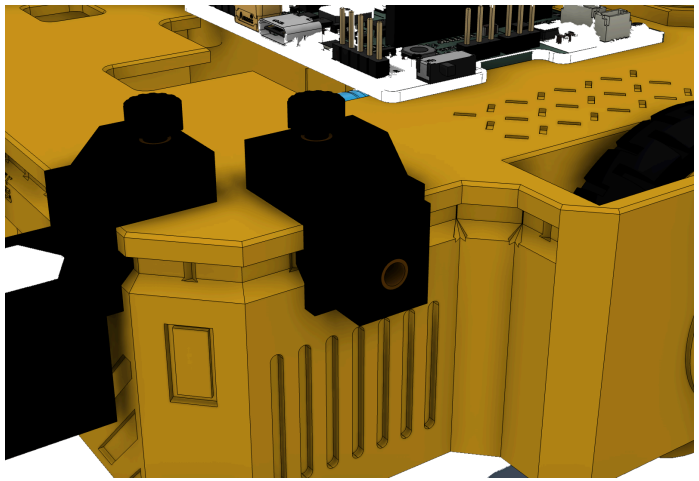
Deck — tread-plate walkways, board window, plug slots



Sonar pod — headlight-style, on the dock plate; the HC-SR04 snaps in from the back



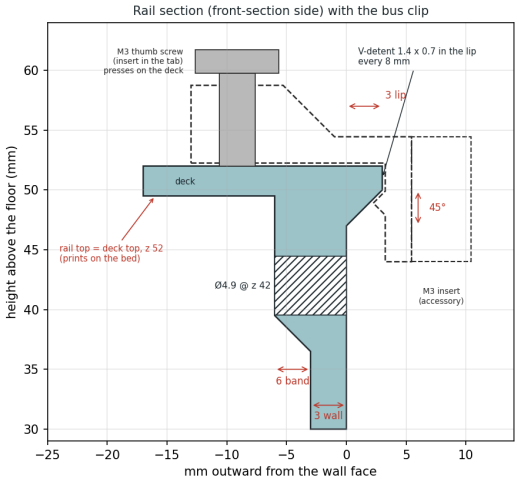
Sonar pod, rear — header up, Dupont plugs inside the pod

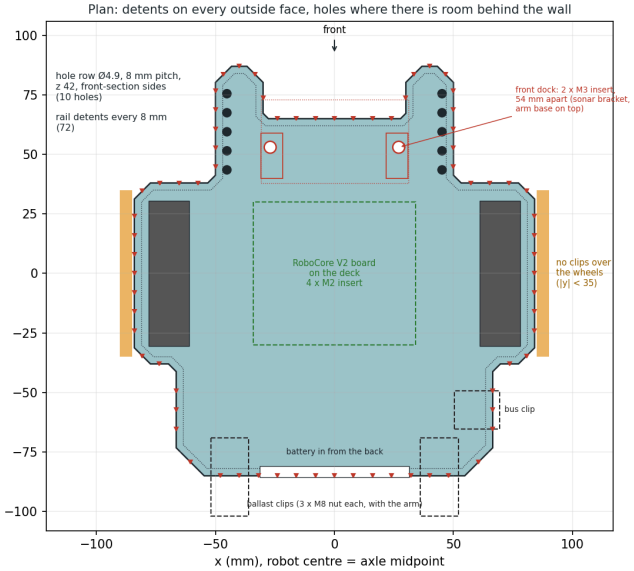


Bus clip — tongue in the groove, thumb screw on the deck, M3 boss for accessories

Expansion bus · rev 5

RoboCore V2 robot, rev 5: mechanical expansion bus





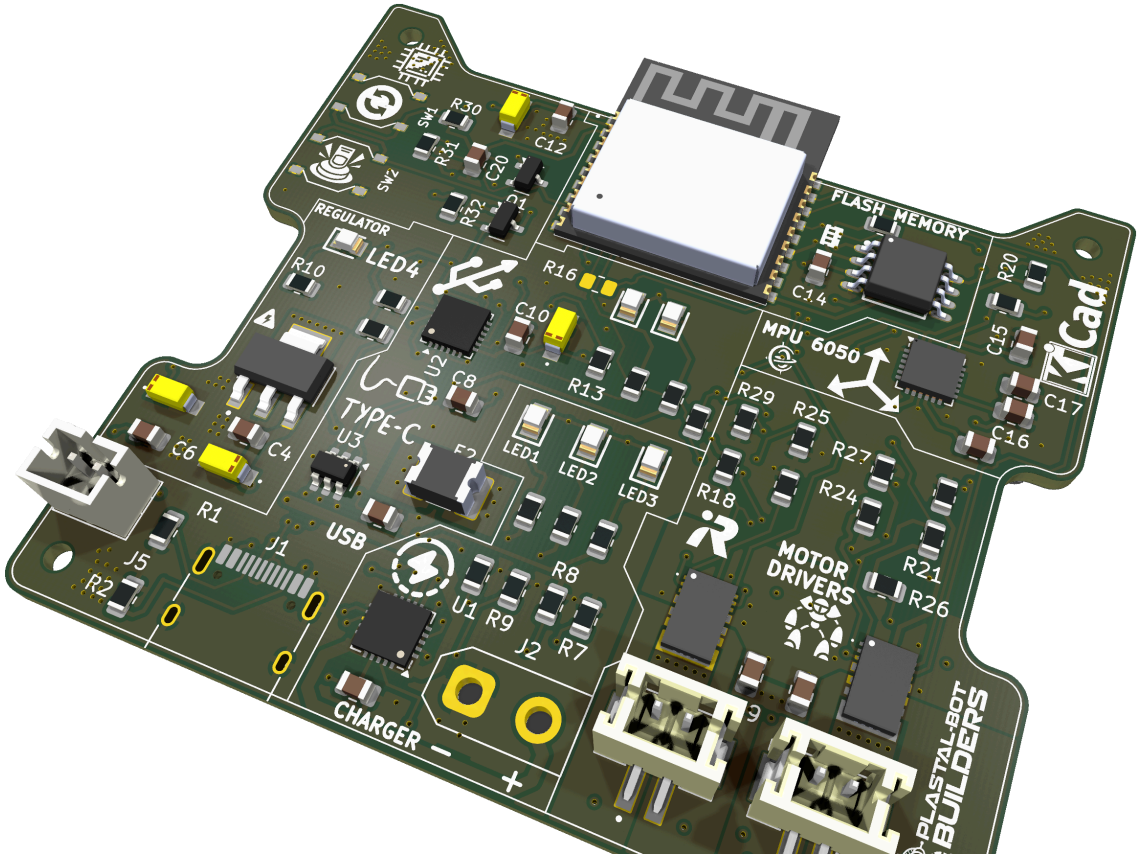
ATTACHMENT POINTS

Feature	Value
Bus groove	1.8 mm deep, back face z 46–49, flat top, 45° underside; every straight run and outside chamfer except the wheel sides
Detents	V-notches 1.4 × 0.7 mm every 8 mm along each run, plus one per chamfer: 54 in all
Bus clip	tongue in the groove + tab on the deck with an M3 insert and an M3 × 8 thumb screw; 10 × 10 mm boss with an M3 insert for the accessory
Hole row	Ø4.9 at z 42, 8 mm pitch, front-section sides only (10 holes); 6 mm wall so Technic pins grip
Front dock	2 × M3 inserts 54 mm apart at (±27, 53), z 52; carries the sonar bracket and the arm module
Deck	z 49.5–52, solid except: board window, USB relief, cable slots, encoder plug slots, rear cable ports, caster vents, wheel wells
Ballast clips	on the rear wall at x ±44, 3 × M8 nuts each; fit them whenever the arm is on
Grid	everything on an 8 mm (LEGO Technic) pitch

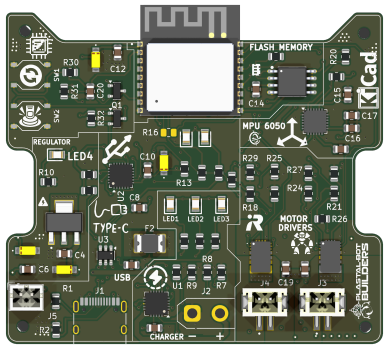
> Print the Rail_coupon first (40 mm of deck, wall and groove, about 20 min). If the clip is tight or loose, change CLIP_GAP (0.25 mm) in fusion/rc_params.py and re-export the clip.

Expansion bus drawing — groove, detents, hole row and front dock; origin at the floor point under the axle middle, x right, y forward, z up (mm)

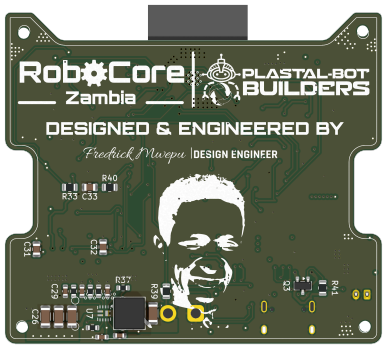
RoboCore V1 · first board, ESP32-C3



RoboCore V1 — ESP32-C3, two DRV8245 motor drivers, MPU-6050 IMU, reworked 2026-10-01 (DRC 0 errors)



top



bottom — boost converter and a few small parts

AT A GLANCE

Item	Value
Board	68 × 60 mm, 2 layers, same outline and holes as V2
MCU	ESP32-C3-WROOM-02 with a CP2102N USB-UART (auto-reset upload)
Motors	2 × TI DRV8245 (one per motor), 6.06 V rail from a TPS61089 boost
IMU	MPU-6050 (obsolete part — replaced by the LSM6DSOX in V2)
Power	1S Li-ion on XT30, MCP73871 USB-C charger (1 A), TLV1117LV33 3.3 V LDO, J5 chassis switch
Extras	W25Q32 SPI flash, battery sense on IO0, artwork on the front

ESP32-C3 PIN MAP

GPIO	Function
IO0	VBAT_SENSE (÷2)
IO1 / IO2	PWM_A / DIR_A
IO3 / IO4	PWM_B / DIR_B
IO5	nFAULT (both drivers)
IO6	nSLEEP (both drivers)
IO7 / IO10 / IO18 / IO19	SPI to the W25Q32 flash
IO8 / IO9	I2C SDA / SCL (MPU-6050); IO9 = BOOT
TXD / RXD	UART to the CP2102N

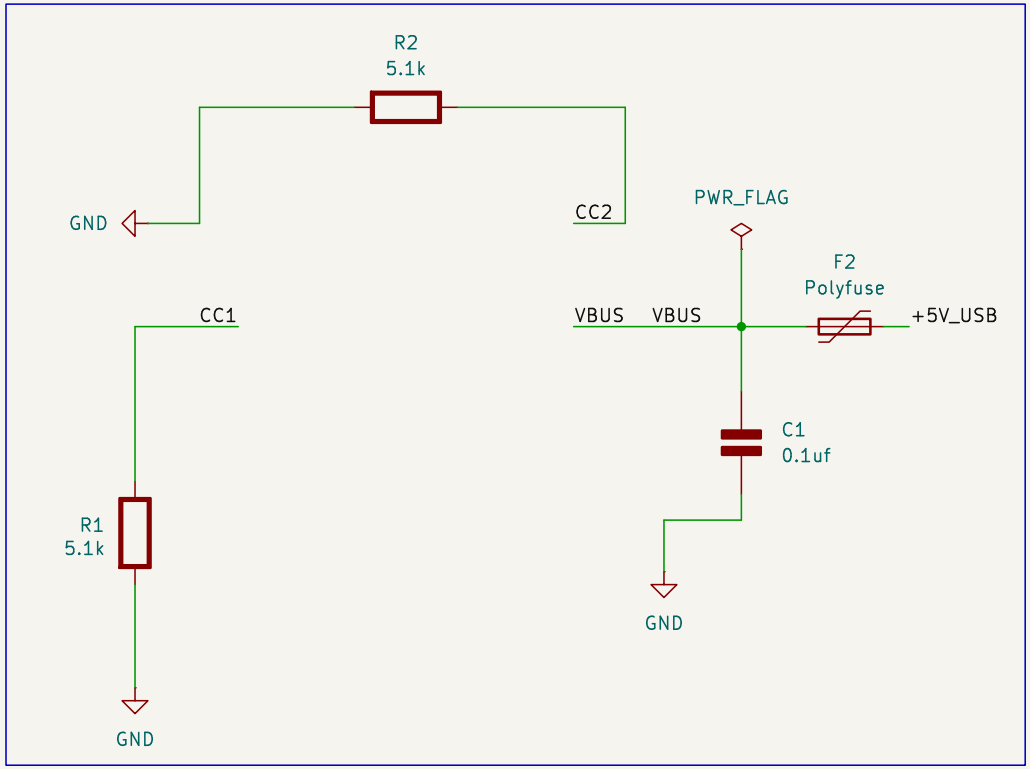
WHAT V2 CHANGED

- ESP32-S3 with native USB (no UART bridge), XRP-compatible pinout and 4 servo headers
- One DRV8411A for both motors, current-limited to 1 A; encoder inputs on the motor sockets
- TPS63001 buck-boost for 3.3 V (works down to a flat cell), 5.5 V servo rail, battery reverse / over-voltage cut-off
- 4 layers, all parts on top, back reserved for artwork

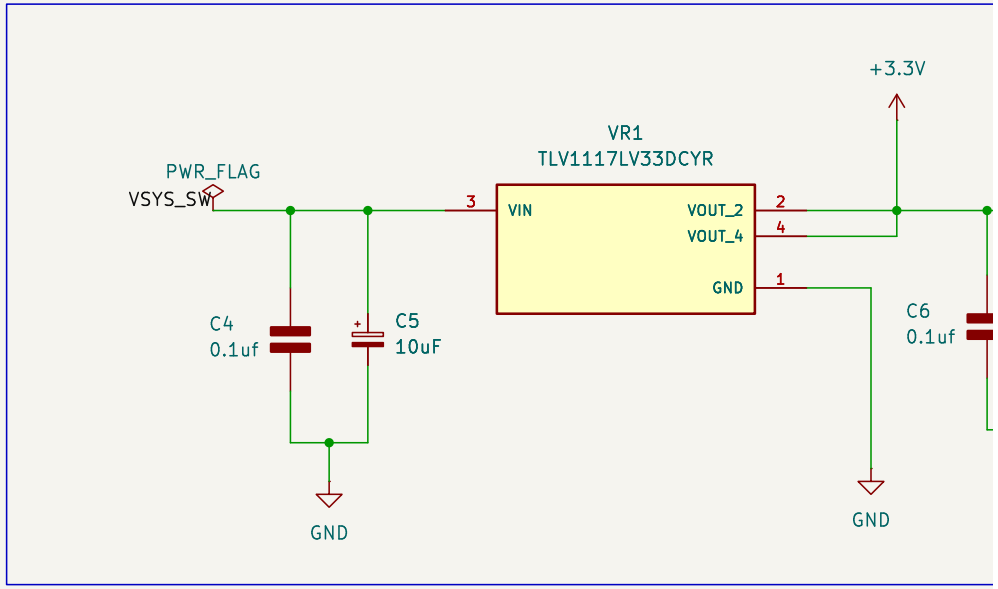
RoboCore V1 schematic · Root sheet · part 1 of 4 (top left)

scale 1.33

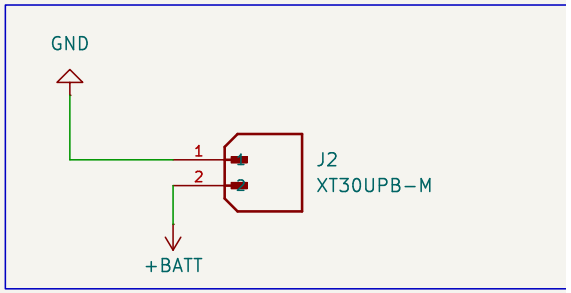
USB–C CONNECTOR



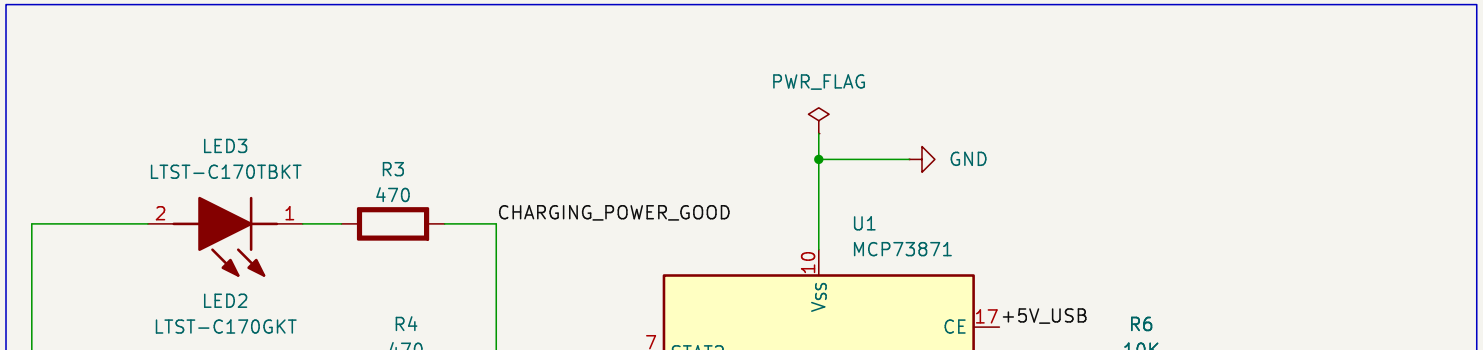
LOW–DROP OUT VOLTAGE REGULATORS



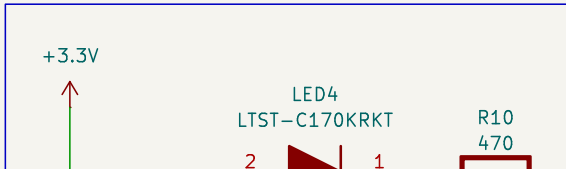
BATTERY CONNECTOR (XT30, 1S Li–ion)

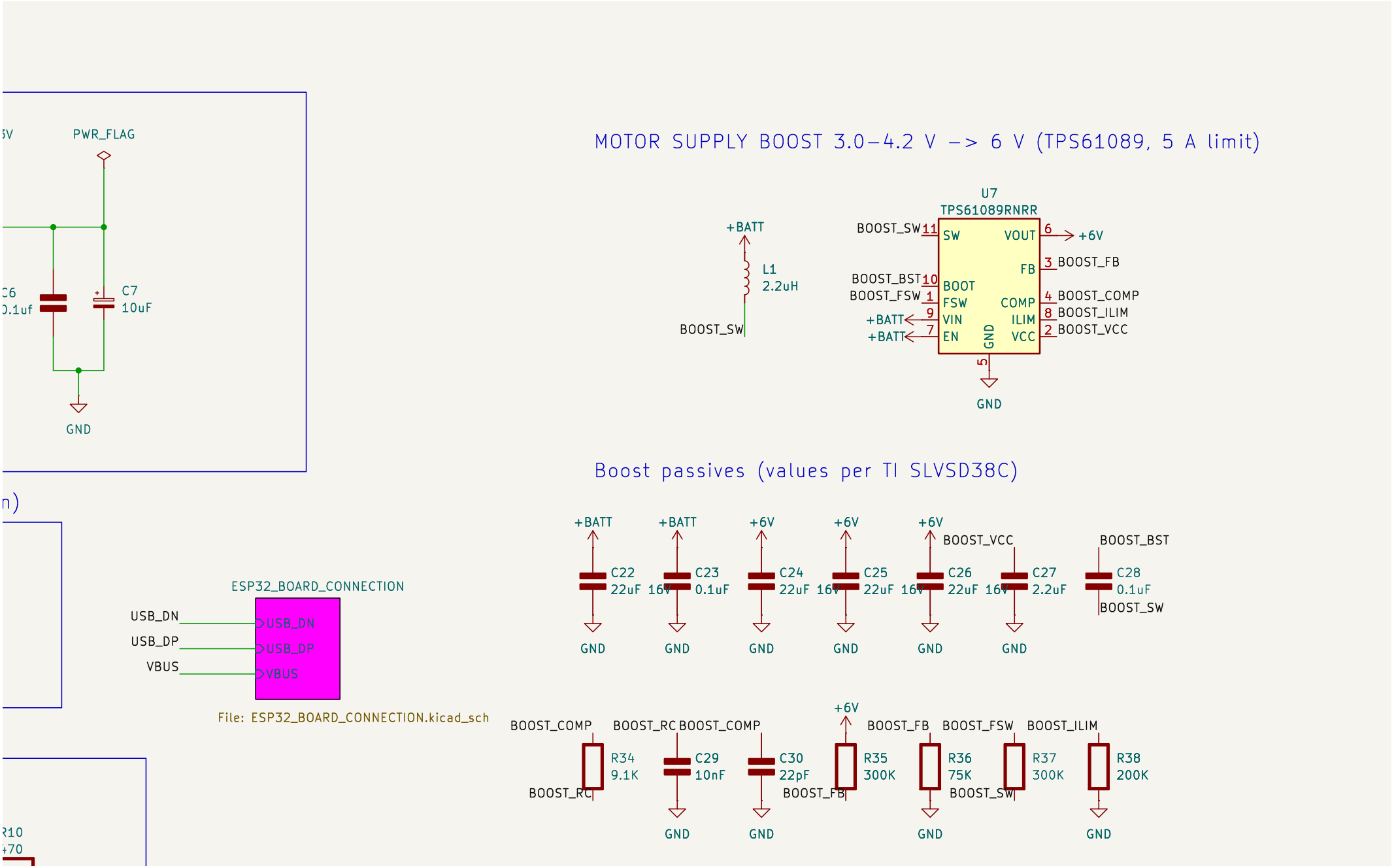


USB BATTERY



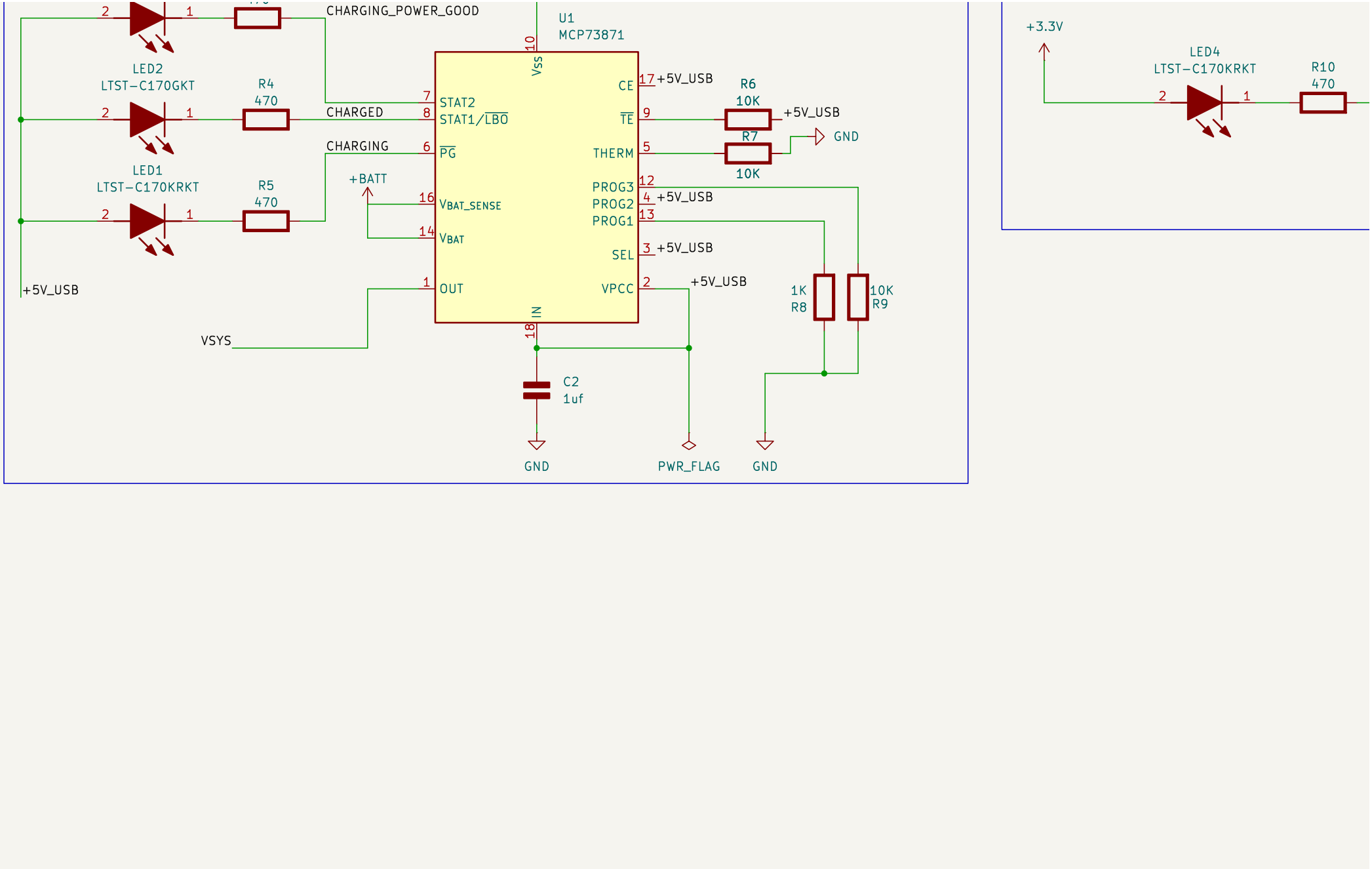
POWER INDICATOR LED




5V
PWR_FLAG
C6
0.1uf
C7
10uF
GND
n)
R10
470
MOTOR SUPPLY BOOST 3.0–4.2 V –> 6 V (TPS61089, 5 A limit)
+BATT
L1
2.2uH
BOOST_SW
U7
TPS61089RNRR
BOOST_SW 11
SW
VOUT 6
+6V
FB 3
BOOST_FB
BOOST_BST 10
BOOST
BOOST_FSW 1
FSW
COMP 4
BOOST_COMP
ILIM 8
BOOST_ILIM
+BATT 9
VIN
+BATT 7
EN
GND
VCC 2
BOOST_VCC
5
GND
Boost passives (values per TI SLVSD38C)
+BATT
C22
22uF 16V
GND
+BATT
C23
0.1uF
GND
+6V
C24
22uF 16V
GND
+6V
C25
22uF 16V
GND
+6V
C26
22uF 16V
GND
BOOST_VCC
C27
2.2uF
GND
BOOST_BST
C28
0.1uF
BOOST_SW
ESP32_BOARD_CONNECTION
USB_DN
USB_DP
VBUS
USB_DN
USB_DP
VBUS
File: ESP32_BOARD_CONNECTION.kicad_sch
BOOST_COMP
R34
9.1K
BOOST_RC
BOOST_RC
C29
10nF
GND
BOOST_COMP
C30
22pF
GND
+6V
R35
300K
BOOST_FB
BOOST_FB
R36
75K
GND
BOOST_FSW
R37
300K
BOOST_SW
BOOST_ILIM
R38
200K
GND

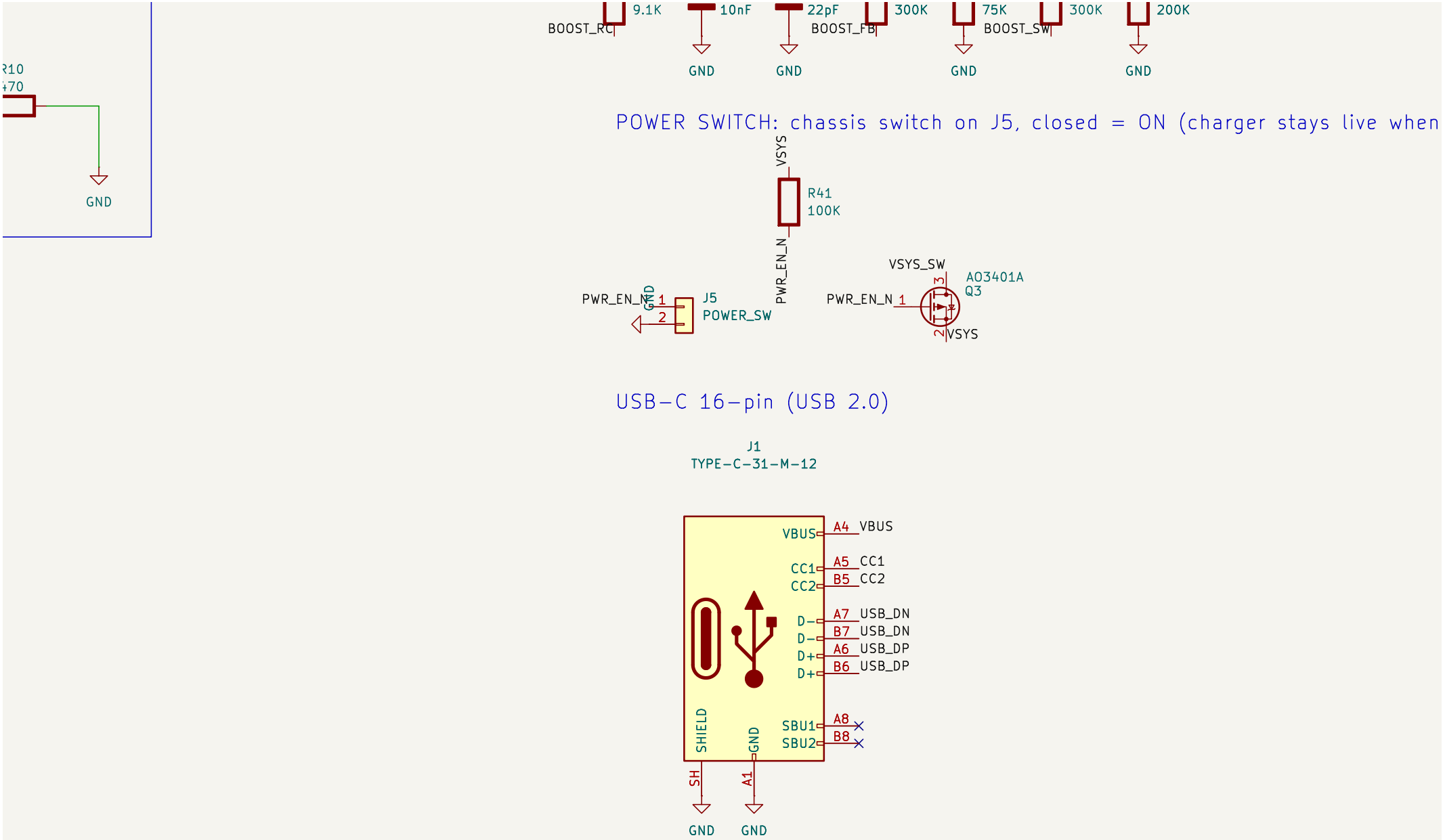
RoboCore V1 schematic · Root sheet · part 3 of 4 (bottom left)

scale 1.33



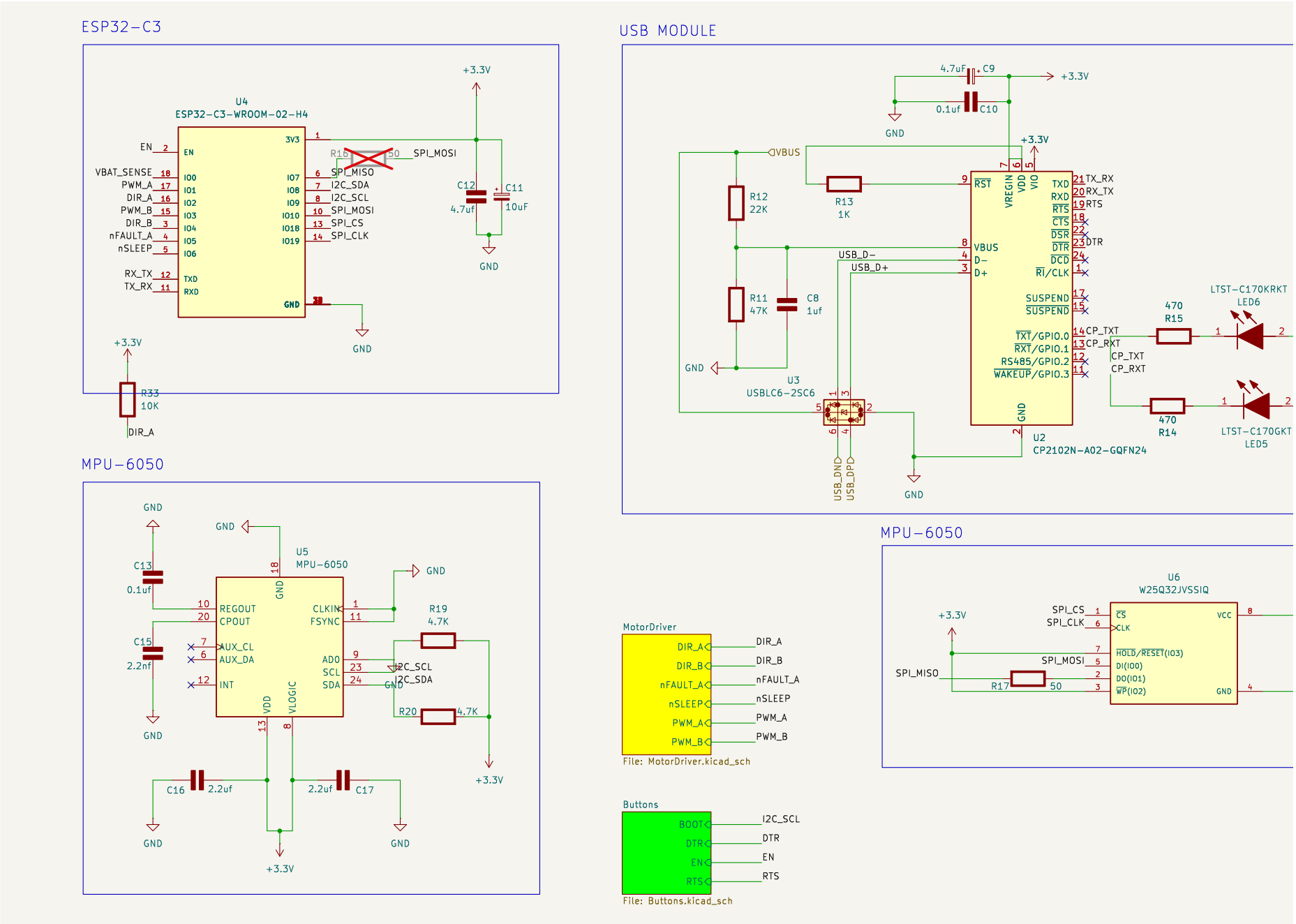
RoboCore V1 schematic · Root sheet · part 4 of 4 (bottom right)

scale 1.33



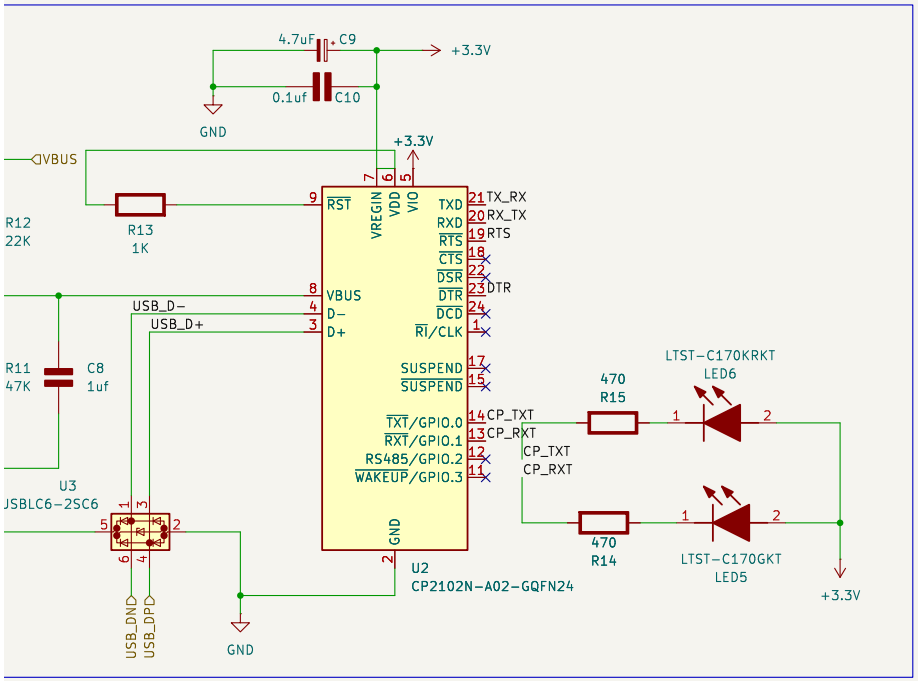
RoboCore V1 schematic · ESP32 board connection · part 1 of 2 (left)

scale 0.95

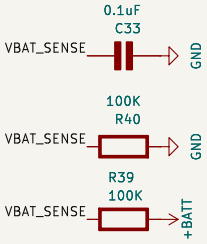


RoboCore V1 schematic · ESP32 board connection · part 2 of 2 (right)

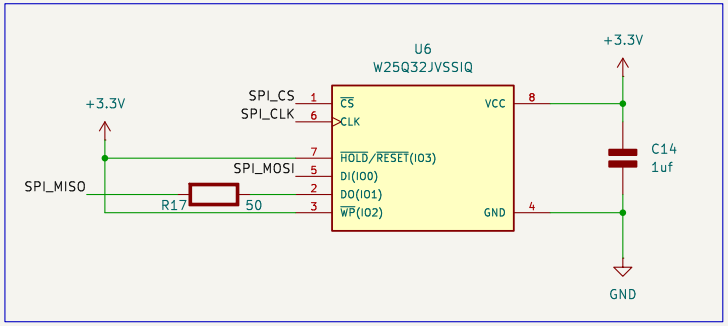
scale 0.95



BATTERY VOLTAGE SENSE -> IO0 (ADC1_CH0), 4.2 V -> 2.1 V



MPU-6050



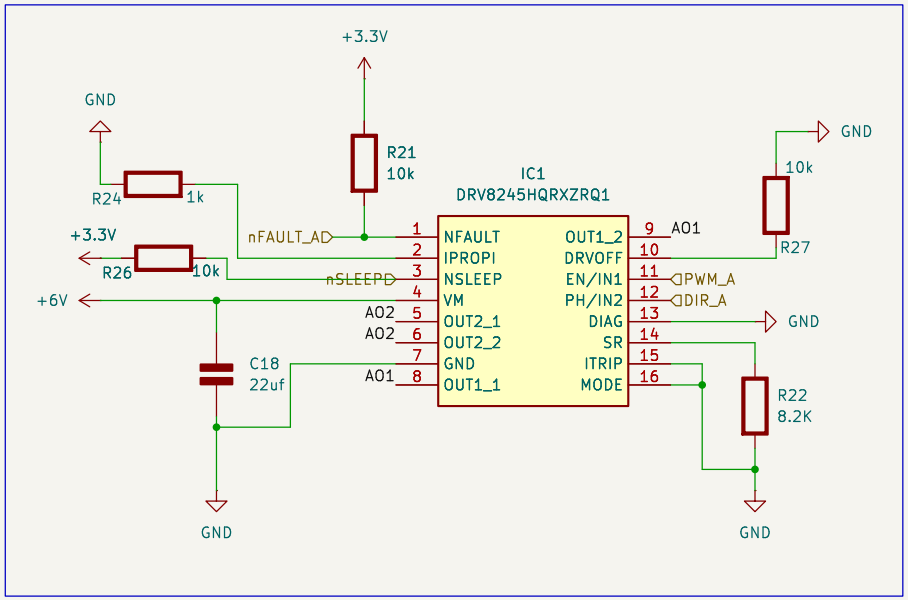
_DIR_A
_DIR_B
_nFAULT_A
_nSLEEP
_PWM_A
_PWM_B

__I2C_SCL
__DTR
__EN
__RTS

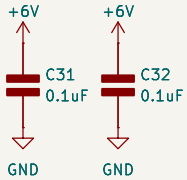
RoboCore V1 schematic · Motor driver

scale 1.11

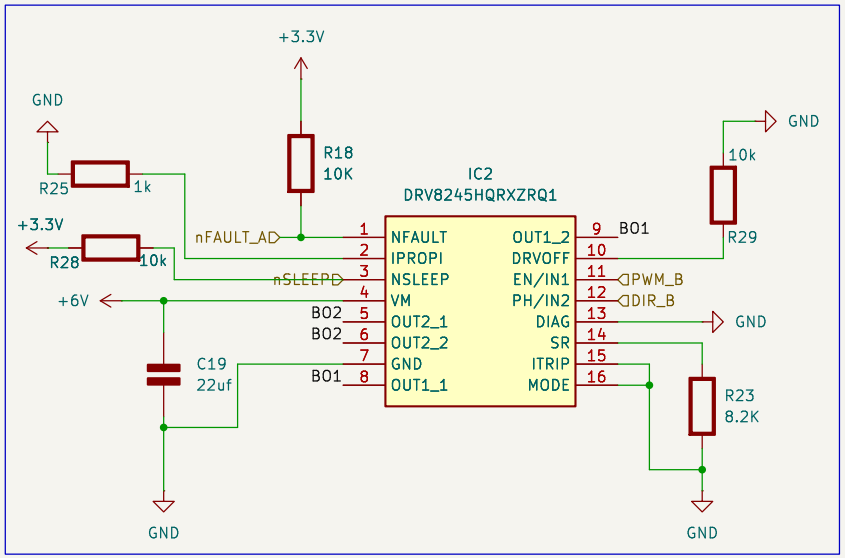
MOTOR A



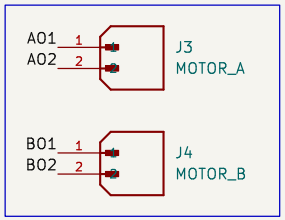
VM decoupling: TI requires 0.1 uF close to each VM pin



MOTOR B



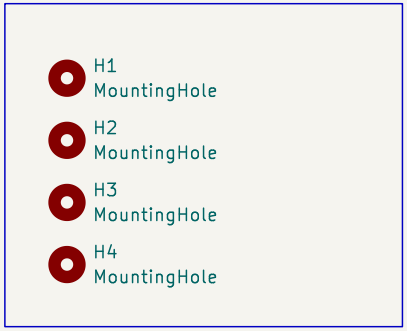
OUTPUT PINS

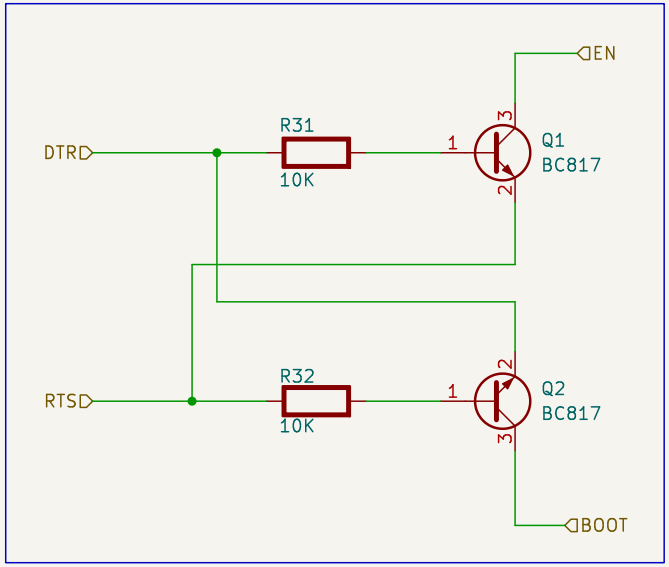


RoboCore V1 schematic · Buttons

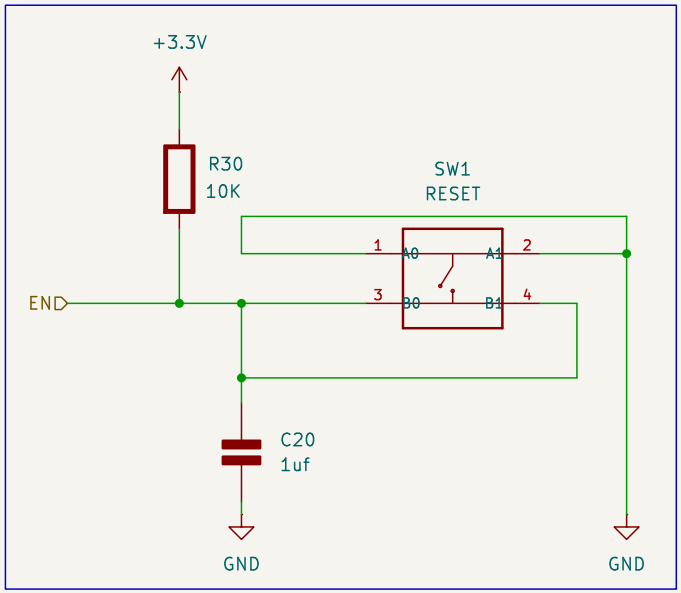
scale 1.30

MOUNTING HOLES





RESET BUTTON



BOOT BUTTON